

ABSTRACT

This thesis presents a multi-agent decision support system for stock trading on the Warsaw Stock Exchange, an underrepresented Central and Eastern European market, and evaluates it under a protocol designed to address four weaknesses that recur in empirical studies of trading strategies: in-sample parameter selection, absent transaction costs, neglect of risk-adjusted performance, and lack of significance tests.

The system consists of five heterogeneous technical-analysis agents, each confined to a single methodological family, feeding a transparent, confidence-weighted decision agent whose aggregation rule is grounded in forecast-combination theory. The evaluation combines anchored walk-forward tuning, calibrated commissions and slippage, the Sharpe, Sortino, Calmar and Martin ratios together with the Ulcer Index, a volatility-matched comparison, bootstrap confidence intervals and paired significance tests, a decomposition of the test window into market regimes, a leave-one-agent-out ablation, sensitivity analyses over the decision-agent weights, the selection objective and transaction costs, and an empirical audit of the confidence heuristic. Two machine-learning baselines, an LSTM directional classifier and a gradient-boosted-trees classifier, are evaluated under the identical protocol.

On 435 held-out trading days for WIG20, the system trails passive buy-and-hold on gross total return in a bull-dominated window, but improves every risk-adjusted metric, halves the maximum drawdown, reduces the integrated drawdown by nearly two-thirds, and overtakes the benchmark on total return once both series are scaled ex ante to the same volatility. These differences are economically meaningful yet not statistically significant on a single sub-two-year window.

Regime analysis attributes the profile to smaller losses during corrections and gains during sideways phases, offset by lower bull-market exposure. The same qualitative pattern appears in an external validation on the FTSE 100 under UK-specific costs, and the system remains competitive with both machine-learning baselines. A dedicated experiment shows that the confidence values carry no measurable information about signal-level accuracy, so the operative mechanism is the pooling of weakly and negatively correlated signals rather than precision weighting. The results support theory-motivated ensembles as risk-aware decision support rather than as return enhancers.

Keywords:

algorithmic trading, decision support systems, multi-agent systems, forecast combination, risk-adjusted performance, walk-forward validation, cross-market validation, Warsaw Stock Exchange.

Field of science and technology in accordance with OECD requirements:

engineering and technology; electrical, electronic and information engineering; computer and information sciences.

STRESZCZENIE

W pracy przedstawiono wieloagentowy system wspomagania decyzji dla handlu akcjami na Giełdzie Papierów Wartościowych w Warszawie, czyli na rynku Europy Środkowo-Wschodniej rzadko analizowanym w literaturze, oraz poddano go ocenie według protokołu usuwającego cztery słabości powtarzające się w badaniach empirycznych strategii inwestycyjnych: dobór parametrów na danych, na których strategia jest oceniana, pominięcie kosztów transakcyjnych, pominięcie miar skorygowanych o ryzyko oraz brak testów istotności statystycznej.

System składa się z pięciu heterogenicznych agentów analizy technicznej, z których każdy wykorzystuje wyłącznie jedną rodzinę metodologiczną wskaźników, oraz z przejrzystego agenta decyzyjnego, który agreguje ich sygnały z wagami wynikającymi z pewności sygnału. Regułę agregacji osadzono w teorii kombinacji prognoz (ang. *forecast combination*). Ocena łączy strojenie parametrów metodą walidacji kroczącej z zakotwiczonym oknem uczącym (ang. *anchored walk-forward*), skalibrowane koszty transakcyjne, miary skorygowane o ryzyko wraz z indeksem Ulcera, porównanie przy zrównanej zmienności, przedziały ufności i sparowane testy istotności metodą bootstrap, dekompozycję okna testowego na reżimy rynkowe, analizę ablacyjną, analizy wrażliwości oraz empiryczną weryfikację heurystyki pewności sygnału. Dwa modele uczenia maszynowego, klasyfikator kierunkowy LSTM oraz klasyfikator oparty na wzmacnianych gradientowo drzewach decyzyjnych, oceniono według identycznego protokołu.

W okresie 435 sesji giełdowych wyłączonych z procesu uczenia system uzyskał dla indeksu WIG20 niższą stopę zwrotu brutto niż pasywna strategia kup i trzymaj w oknie zdominowanym przez trend wzrostowy, poprawił jednak każdą z rozważanych miar skorygowanych o ryzyko, zmniejszył o połowę maksymalne obsunięcie kapitału, ograniczył skumulowane obsunięcie o niemal dwie trzecie oraz przewyższył strategię odniesienia pod względem stopy zwrotu po sprowadzeniu obu szeregów do tej samej zmienności wyznaczonej z góry. Różnice te są istotne ekonomicznie, lecz nie osiągają istotności statystycznej w oknie krótszym niż dwa lata.

Analiza reżimów rynkowych wskazuje, że profil ten wynika z mniejszych strat w okresach korekty i zysków w okresach konsolidacji, okupionych mniejszym udziałem w fazach wzrostowych. Ten sam wzorec potwierdziła walidacja zewnętrzna na indeksie FTSE 100, a system pozostał konkurencyjny wobec obu modeli uczenia maszynowego. Odrębny eksperyment wykazał, że wartości pewności sygnału nie niosą mierzalnej informacji o trafności pojedynczych sygnałów, zatem mechanizmem działania systemu jest łączenie sygnałów słabo i ujemnie skorelowanych, a nie wążenie precyzją. Wyniki potwierdzają przydatność zespołów agentów jako narzędzia wspomagania decyzji świadomego ryzyka, a nie metody zwiększania stopy zwrotu.

Słowa kluczowe:

handel algorytmiczny, systemy wspomagania decyzji, systemy wieloagentowe, kombinacja prognoz, efektywność skorygowana o ryzyko, walidacja krocząca, walidacja międzyrynkowa, Giełda Papierów Wartościowych w Warszawie.

Dziedzina nauki i techniki, zgodnie z wymogami OECD:

nauki inżynieryjne i techniczne; elektrotechnika, elektronika, inżynieria informatyczna; nauki o komputerach i informatyka.

LIST OF IMPORTANT SYMBOLS AND ABBREVIATIONS

Symbols of the forecast-combination model

f_k	$[\cdot]$	–	forecast produced by the k -th forecaster
e_k	$[\cdot]$	–	forecast error of the k -th forecaster, $e_k = y - f_k$
σ_k^2	$[\cdot]$	–	variance of the forecast error of the k -th forecaster
$\rho_{k,j}$	$[\cdot]$	–	correlation between the errors of forecasters k and j
w_k	$[\cdot]$	–	weight assigned to the k -th forecaster in the combination
K	$[\cdot]$	–	number of forecasters in the pool
y	$[\cdot]$	–	unobserved target of the forecast

Symbols of the decision layer

$\sigma_{a,s}(t)$	$[\cdot]$	–	directional signal of agent a for symbol s at date t , $\in \{-1, 0, +1\}$
$c_{a,s}(t)$	$[\cdot]$	–	confidence of agent a for symbol s at date t , $\in [0, 1]$
$\kappa(a)$	$[\cdot]$	–	methodological family of agent a
w_κ	$[\cdot]$	–	family-level weight of family κ
$c_{\min}$	$[\cdot]$	–	minimum confidence below which a signal is discarded
$\theta_{\text{buy}}, \theta_{\text{sell}}$	$[\cdot]$	–	decision thresholds of the decision agent
m	$[\cdot]$	–	decision-agent mode: passive, balanced or aggressive

Symbols of the indicators

h	$[\cdot]$	–	value of the MACD histogram
σ	$[\cdot]$	–	standard deviation over a rolling window
τ	$[\cdot]$	–	threshold of the rate-of-change rule
U, L	$[\cdot]$	–	upper and lower Bollinger band
m	$[\cdot]$	–	Bollinger band multiplier
RS	$[\cdot]$	–	relative strength, the ratio of average gain to average loss

Symbols of the evaluation

r_t	$[\cdot]$	– portfolio return on day t
$\bar{r}$	$[\cdot]$	– mean daily return
N	$[\cdot]$	– number of observations in the evaluation window
dd_t	$[\cdot]$	– running drawdown of the equity curve at date t
λ	$[\cdot]$	– ex-ante leverage factor of the volatility-matched comparison
σ^{target}	$[\cdot]$	– target annualised volatility of the benchmark
σ^{strat}	$[\cdot]$	– annualised volatility of the strategy on the training window
γ	[bps]	– proportional commission rate
s	[bps]	– one-way slippage rate
$T_{\text{tr}}, T_{\text{te}}$	$[\cdot]$	– training and test windows
ρ	$[\cdot]$	– Spearman rank correlation coefficient
δ	$[\cdot]$	– half-width of the no-trade band of a machine-learning baseline
h	[d]	– prediction horizon of a machine-learning baseline
n	$[\cdot]$	– number of pooled signal observations

Abbreviations used in the thesis

B&H	– buy-and-hold, the passive benchmark strategy
bps	– basis point, one hundredth of a percentage point
CAGR	– compound annual growth rate
CEE	– Central and Eastern Europe
CI	– confidence interval
CSV	– comma-separated values, the price-data file format
DSS	– decision support system
EMA	– exponential moving average
FTSE 100	– Financial Times Stock Exchange 100 Index (London Stock Exchange)
GBM	– gradient boosting machine, the gradient-boosted-trees baseline
LSTM	– long short-term memory, a recurrent neural-network architecture
MACD	– moving-average convergence-divergence
MAS	– multi-agent system
MDD	– maximum drawdown
OHLCV	– open, high, low, close and volume price data
ROC	– rate of change
RSI	– relative strength index
SDRT	– Stamp Duty Reserve Tax, the UK buy-side transaction tax
SLR	– systematic literature review
SMA	– simple moving average
TR	– total return
UI	– Ulcer Index
UML	– unified modeling language
WIG20	– Warszawski Indeks Gieldowy 20, the WSE blue-chip index
WSE	– Warsaw Stock Exchange
YAML	– YAML Ain't Markup Language, the configuration file format

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1. INTRODUCTION

Modern financial markets run substantially on decision support systems and algorithmic trading. Institutional and retail participants alike depend on them to reduce large volumes of market information to something that can be acted upon [1]. What that machinery consists of has changed considerably over two decades of research. Rule-based strategies came first [2], classifiers drawn from machine learning followed [3, 4], and the most recent generation learns its trading policy from the data directly, by deep reinforcement learning [5, 6, 7].

Running alongside that progression, multi-agent systems (MAS) have been put forward as a way of imposing order on so many competing rules and models: independent agents each issue a signal, and a coordinator resolves those signals into one decision [8, 9, 10, 11]. What recommends the arrangement is the separation it enforces between two questions that are easily run together, namely what a given indicator reports about the market and what a portfolio ought to do in response.

The practical problem that motivates this thesis is narrower than the breadth of that literature suggests. A retail investor on the Warsaw Stock Exchange (WSE) has access to daily price data, a modest brokerage account, and a large menu of published trading rules, but has no reliable way to tell which of those rules, or which combination of them, is worth following. The published evidence is not much help, because a substantial part of it is produced under evaluation protocols that make results look better than they are.

1.1. Problem statement

Empirical studies of trading strategies are prone to four methodological weaknesses, and in combination these undermine the credibility of what such studies report.

In-sample parameter selection is the first. Choosing the best configuration of a strategy on the very data that will afterwards be used to judge it inflates the apparent out-of-sample performance [12, 13]. *Omission of realistic frictions* is the second, commissions and slippage above all, which between them can consume a noticeable share of whatever gross return a high-turnover configuration produces.

Neglect of risk-adjusted performance is the third. Cumulative profit alone is a common basis for setting a trading strategy against passive buy-and-hold, and it leaves out of account that an active strategy may deliver a smoother equity curve while returning less unleveraged. Sustained bull markets sharpen the problem, since anything less than continuous exposure is mechanically outpaced there by a fully invested benchmark [14]. *Absence of formal significance tests* is the fourth. Headline figures appear as bare point estimates, unaccompanied by confidence intervals or by paired tests against the benchmark, notwithstanding that the metrics involved are known to be non-Gaussian and path-dependent [15, 12].

A geographical concentration compounds these methodological ones. Evidence on multi-agent decision support systems clusters heavily on the major markets of the United States, Western Europe and Asia, whereas the Warsaw Stock Exchange, the largest in Central and Eastern Europe (CEE) by capitalisation, has drawn comparatively little study [16, 17, 18]. The difficulty facing a retail investor there is consequently twofold: the evidence available is methodologically fragile, and such of it as exists was collected somewhere else.

1.2. Aim and scope of the thesis

The aim of this thesis is to design a transparent multi-agent decision support system for trading large-capitalisation stocks on the Warsaw Stock Exchange, and to determine, under an evaluation protocol that addresses the four weaknesses stated above, whether such a system offers a measurable advantage over passive investment.

The scope is defined by four decisions. The system trades daily closing prices of WIG20 index constituents, so intraday behaviour, order-book dynamics, and other asset classes are outside its remit. The signal agents use only price-derived technical indicators, so fundamental and alternative data are excluded. The aggregation rule is a linear, confidence-weighted score, chosen so that the decision layer remains interpretable and its mapping to forecast-combination theory stays transparent, in contrast to opaque end-to-end learners. Finally, all evidence is produced by simulation on historical data, so no live-trading exercise is part of the work.

Three hypotheses, stated formally in Section 3.3 and tested in Chapter 5, give the aim its testable form. In summary, they assert that the system matches or exceeds passive buy-and-hold on risk-adjusted metrics even where it does not on gross return (H1), that it outperforms the best single-agent baseline on total return (H2), and that its performance is robust to the removal of any single agent (H3).

The technical assumptions adopted throughout are the following. Positions are opened and closed at the daily closing price. Transaction costs comprise a proportional commission and one-way slippage, calibrated per market. Only decision-agent parameters are tuned, and only on a training window that precedes the evaluation window chronologically; all indicator hyperparameters are fixed at their literature-conventional values. Portfolio capital is allocated as a fixed fraction of available cash per position, without leverage.

1.3. Own contribution

The work reported in this thesis is the author's own. The system, the experiments, and the analysis were designed, implemented, and carried out by the author. Specifically, the author's contribution comprises the following:

- the design and implementation of the complete software system described in Chapter 4, comprising five signal agents, the decision agent, the portfolio simulator, the metrics calculator, and the experiment harness,
- the design of the evaluation protocol of Chapter 5, including the anchored walk-forward procedure, the transaction-cost calibration, the market-regime labelling, the volatility-matched comparison, and the bootstrap testing scheme,
- the implementation and in-sample tuning of the two machine-learning baselines used as reference points,
- the conceptual grounding of the aggregation rule in forecast-combination theory, together with the correlation and confidence-validation experiments that test how far that grounding holds empirically,
- the execution and interpretation of all experiments reported in Chapter 5, including the external validation on the FTSE 100.

1.4. Structure of the thesis

The thesis is organised into six chapters, whose significance is unequal by design: Chapters 4 and 5 carry the original work, while the remaining chapters frame it.

Chapter 2 reviews the state of the art across four streams: technical analysis and algorithmic trading; multi-agent systems in finance; the algorithmic trading literature on Central and Eastern European markets; and the evaluation of risk-adjusted performance. It establishes both the methodological gap and the empirical niche that the thesis addresses.

Chapter 3 presents the theoretical background and the system design. It introduces forecast-combination theory and the variance-reduction argument that motivates the architecture, maps that theory onto the trading problem, states the three hypotheses, and then describes the four-layer architecture, the five signal agents, the decision agent, and the agent taxonomy.

Chapter 4 describes the software realisation: the requirements it had to satisfy, the technologies selected and why, the structure of the code, the data preparation pipeline, and the way the implementation was verified.

Chapter 5 reports the empirical part of the work. It describes the evaluation protocol in full, reports the results on the WIG20 panel, subjects them to a battery of robustness checks and an external validation on the FTSE 100, and interprets the evidence against the three hypotheses.

Chapter 6 summarises what was achieved, states the conclusions as an organised list, discusses the limitations of the study and the difficulties encountered during the work, and indicates directions for further development.

2. STATE OF THE ART

Four streams of literature bear on the work reported in this thesis and are reviewed below: technical analysis and algorithmic trading; multi-agent systems in finance; the algorithmic trading literature on Central and Eastern European markets; and the evaluation of risk-adjusted performance. Dividing the field this way locates the thesis within a crowded research landscape and shows why it is the conjunction of the four, rather than any one of them alone, that leaves a gap to be addressed.

2.1. Review method

The literature in this chapter was assembled in two ways, and the distinction matters for what the chapter can claim. The two streams that carry the empirical argument of the thesis, namely multi-agent systems for trading decision support and algorithmic trading on Central and Eastern European markets, were surveyed through a structured database search. The two remaining streams, technical analysis and risk-adjusted performance evaluation, were not searched systematically: they supply the theoretical apparatus and the measurement instruments used later in the thesis, and are cited as such.

The structured search addressed two questions:

RQ1 What multi-agent or ensemble architectures have been proposed for trading or investment decision support in financial markets, and under what evaluation protocols have they been assessed?

RQ2 What algorithmic trading and market-efficiency research exists for the Warsaw Stock Exchange and other Central and Eastern European equity markets?

RQ1 establishes what has already been built, which is what allows the thesis to locate its contribution in the evaluation rather than in the aggregation rule. RQ2 tests the claim that the Warsaw Stock Exchange is underrepresented in this literature. Neither claim can be supported by selective reading, which is why these two streams, and only these two, were searched.

Three databases were searched in February 2026: Scopus, Web of Science and IEEE Xplore. Queries were restricted to English-language journal articles, conference papers and peer-reviewed magazine articles published between 2000 and 2026. Listings 2.1 and 2.2 give the two queries in a database-independent form; each was translated into the query syntax of the database concerned, so the three executed forms are equivalent in concept structure but differ in notation. The verbatim strings, per-database counts and run dates are archived with the thesis materials.

Duplicates were removed on digital object identifier, falling back to normalised title for the records that carry none; author names were not used for matching, because the three databases render forenames and diacritics inconsistently and author-based matching fails silently on precisely the records where a missed duplicate matters. Table 2.1 gives the yield by database.

```

( "multi-agent" OR "multiagent" OR "agent-based"
OR "trading agent" OR "software agent" )
AND
( "stock market" OR "stock trading" OR "stock exchange" OR "equity market"
OR "algorithmic trading" OR "automated trading" OR "portfolio management"
OR "financial market" OR "foreign exchange market" OR "cryptocurrency" )
AND
( "decision support" OR "trading strategy" OR "trading system"
OR "trading decision" OR "trading signal" OR "investment decision"
OR "market prediction" OR "price prediction" OR "forecast*"
OR "reinforcement learning" )
AND NOT
( "electricity market" OR "energy trading" OR "smart grid"
OR "power system" OR "supply chain management" )

```

Listing 2.1. Query for RQ1, in database-independent form. The three concept blocks are architecture, financial domain and function; the negation block removes the energy-trading literature, in which *multi-agent trading* is a common term for a different subject.

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( "Warsaw Stock Exchange" OR "WIG20" OR "Polish stock market" OR "GPW"
OR ( ( "Poland" OR "Polish" OR "Central and Eastern Europe" OR "Visegrad" )
AND
( "stock market" OR "stock exchange" OR "equity market"
OR "capital market" ) ) )
AND
( "algorithmic trading" OR "automated trading" OR "trading strategy"
OR "technical analysis" OR "trading rule" OR "market efficiency"
OR "stock market prediction" OR "decision support" OR "trading system"
OR "multi-agent" )

```

Listing 2.2. Query for RQ2, in database-independent form. The specific venue terms match on their own, while the generic country and region terms must co-occur with an equity-market term; without that restriction the query returned a large volume of country-level macroeconomic work.

Table. 2.1. Records returned by each query in each database.

Database	Multi-agent systems	CEE markets	Total
Scopus	705	86	791
Web of Science	339	83	422
IEEE Xplore	279	26	305
Records identified	1,323	195	1,518
Duplicates removed	—	—	449
Unique records	—	—	1,069

Relevance is defined by the criteria below. A study qualifies if it satisfies all of the inclusion criteria and none of the exclusion criteria.

- IC1 it is a peer-reviewed journal article, conference paper, or peer-reviewed technical article in a magazine venue;
- IC2 it was published between 2000 and 2026, in English;
- IC3 it presents or evaluates a computational system or strategy that produces trading or investment decisions;
- IC4 it uses a multi-agent or explicit signal-combination architecture (RQ1), or concerns a Polish or other Central and Eastern European equity market (RQ2);
- IC5 it reports an empirical evaluation on real market data.
- EC1 the subject is energy, electricity, commodity-grid or supply-chain trading, which is a different domain;
- EC2 no empirical evaluation is reported, as in position papers and pure architecture proposals;
- EC3 it is an artificial-market simulation with no evaluated trading decision rule;
- EC4 the full text is not retrievable through institutional access;
- EC5 it is a secondary study; reviews and surveys are used as sources for citation chasing rather than counted as primary studies;
- EC6 it is non-peer-reviewed magazine content, such as an editorial, news item, column, interview or book review.

IC1 and EC6 operate as a pair. Magazine venues carry peer-reviewed technical articles alongside material that is not peer reviewed, so admitting such venues without a criterion on the item type would contradict IC1.

Before screening, the queries were checked against a set of publications known in advance to be relevant, which a correct query must return. All seven were retrieved [8, 10, 11, 6, 5, 17, 18]. The check was worth performing: it revealed that IEEE Xplore classifies *IEEE Network* as a magazine and had therefore filtered out [8] under a document-type restriction that Scopus and Web of Science, which classify the same article as a journal article, did not apply. The restriction was harmonised across the three databases and the affected searches repeated. The episode is worth recording because it shows that a document-type filter does not mean the same thing in every database.

Provenance of the reference set

Because only two of the four streams were searched, the references in this chapter did not all enter by the same route. Three routes are distinguished. The first is the structured search described above. The second is backward and forward citation chasing from the studies it returned; [9] is the clearest case, since its title contains no agent terminology and no abstract is indexed for it, so no keyword query can retrieve it. The third comprises works cited because the thesis *uses* them rather than because it reviews them: the risk-adjusted performance measures [19, 20, 21, 22], the forecast-combination results that motivate the architecture [23, 24, 25], the statistical procedures

applied in Chapter 5 [15, 26, 27], and the machine-learning baselines [28, 29]. Works in this third group were located from the method the thesis needed rather than from a topic query, and a database search is not the appropriate instrument for them: the definition of a Sharpe ratio is not a research finding to be surveyed. A quarter of the references in this chapter predate the year 2000 and therefore fall outside the search window by construction.

2.2. Algorithmic trading and technical analysis

Trading signals in technical analysis are generated from indicators computed over price and volume. That simple rules built on moving averages and trading ranges carry predictive power difficult to square with the efficient markets hypothesis in its strict form was established early [2]. The debate was subsequently put on a firmer footing by algorithmic backtests and statistical testing [30], and indicators as widely used as MACD, RSI and Bollinger Bands were shown to remain informative once paired with machine-learning classifiers [3].

One strand bears especially closely on the ensemble design studied here. Empirical work on momentum, both cross-sectional [31, 32] and in the time series [33], has established that trend-following and mean-reversion signals hold orthogonal information about future returns. Section 3.7 builds its agent taxonomy on precisely that orthogonality.

Attention has since moved to deep learning, to recurrent architectures for predicting price in particular [4], and onward to deep reinforcement learning systems that acquire a trading policy end to end [5, 6, 7, 34]. Reviewing the field systematically yields a mixed verdict on technical analysis, one that turns heavily on which market, which period and which evaluation protocol a given study happens to adopt [35, 36].

Situating a transparent forecast-combination design against that learning-based stream calls for direct comparison, so two learned baselines are carried through the evaluation under the walk-forward protocol, the transaction costs and the portfolio simulator that the ensemble itself receives. One is a directional LSTM classifier after [4]; the other a gradient-boosted-trees classifier [28, 29], drawn from a family that large-scale comparisons place among the strongest learners available for tabular financial features [37, 38].

Two methodological criticisms recur throughout. Backtest overfitting is the first: evaluate enough configurations on one dataset and the winner is biased upward, its out-of-sample performance correspondingly overstated [12, 13]. The second is that realistic frictions frequently go missing, commissions and slippage among them, though together they can consume a large share of the gross return of any high-turnover strategy. Section 5.1 builds both criticisms into the evaluation design rather than setting them aside.

2.3. Multi-agent systems in finance

Trading lends itself naturally to a multi-agent abstraction, since each distinct strategy can be packaged as an independent agent communicating across a well-defined interface. Artificial markets built on that basis have served to study how liquidity, volatility and price discovery emerge from the interaction of heterogeneous traders [39, 40].

Proposals for multi-agent decision support in stock and currency markets go back two decades, and share a common shape: specialised agents issue buy and sell recommendations, and a coordinating component aggregates what they issue [8, 9, 10, 11]. Among these, A-Trader [9, 10] stands closest to the design presented here, resting as it does on several specialised agents together with a consensus-based coordinator, though its evaluation has been conducted mainly

on FOREX data. The paradigm has more recently been extended by multi-agent reinforcement learning, applied to portfolio management [5] and to algorithmic trading [6].

A substantial body of control-theory research outside finance takes up the question of how interacting agents reach agreement under consensus protocols. It covers consensus tracking across networks that mix cooperation with competition [41], rates of convergence when communication is delayed [42], and logic-based switching for uncertain nonlinear multi-agent dynamics [43]. What that work analyses is the dynamical convergence of agent states towards a common value as time passes. The coordinator studied here is something else, a static forecast-combination rule aggregating heterogeneous directional forecasts on each date. Related in spirit the two notions of agreement may be, but formally they are distinct.

Set against that literature, no novelty is claimed for the aggregation mechanism itself, weighted-score combination of specialised agents being well established in the systems cited above. Two things distinguish the present work instead. The first is that it draws the connection between the aggregator and forecast-combination theory [23, 24, 25] explicitly, reading the coordinator score as a weighted sum of heterogeneous directional forecasts. That reading shapes the design of the decision agent and motivates the ablation and correlation experiments alike, and its limits are established empirically in Sections 5.2.8 and 5.3.2, which validate the confidence values and probe how far the weights matter.

The second, and the weightier of the two, is that the resulting ensemble is put through the rigorous evaluation protocol of Section 5.1, whose elements most prior multi-agent trading studies do not apply together. Theoretical grounding and evaluation are therefore where the intended contribution lies, and not in the aggregation rule.

2.4. Algorithmic trading on Central and Eastern European markets

Size has not translated into attention. Measured by capitalisation no market in Central and Eastern Europe is larger than the Warsaw Stock Exchange, yet the algorithmic trading literature has given it little space. What study the WSE has received has gone to the co-movement of the Polish market with others and the macroeconomic drivers behind it [16], to tests of the informational efficiency of the exchange [17], and to the interaction of algorithmic trading with liquidity and volatility among blue chips [18].

The author knows of no study evaluating a multi-agent decision support system over WIG20 that combines an explicit walk-forward protocol with realistic costs and a comparison on the risk axis. That absence is the primary empirical niche this thesis occupies. Whether the profile obtained belongs to WSE conditions specifically is settled afterwards, by rebuilding the pipeline in full on the FTSE 100 with costs calibrated to the United Kingdom (Section 5.3.7).

2.5. Risk-adjusted performance evaluation

A literature of its own has grown up around risk-adjusted measurement, extending well past the Sharpe ratio [19]. Penalising downside deviation alone, as the Sortino ratio [20] does, sits closer to the way losses are actually experienced; setting annualised return against maximum drawdown, as the Calmar ratio [21] does, captures the worst loss realised anywhere along the equity path. The Ulcer Index and the Martin ratio [22] carry the drawdown-based approach further still, taking in the whole history of drawdowns instead of the peak-to-trough extremum alone.

On the adaptive markets hypothesis [44], efficiency is not a fixed property of a market but varies over time, which supplies a theoretical reason to assess a strategy regime by regime rather

than on aggregates alone. The evaluation here follows that reasoning, reporting the standard risk-adjusted ratios and decomposing performance across distinct market regimes, and comparing the equity curve of the ensemble against the benchmark once the two have been matched on ex-ante volatility. That risk-aware protocol is among the methodological contributions of this thesis.

2.6. Chapter summary

The four streams reviewed above converge on a single observation. Technical trading rules and multi-agent architectures for combining them are both well established, but the empirical evidence about them is fragile, because it is frequently produced under evaluation protocols that select parameters in sample, ignore transaction costs, report only cumulative profit, and omit significance tests. The risk-adjusted evaluation literature supplies the metrics needed to repair the third of these weaknesses, and the forecast-combination literature supplies a theory that explains what a coordinator of heterogeneous agents should be expected to achieve. What is missing is a study that brings these elements together on a market outside the small set of heavily studied exchanges. That is the gap the remainder of this thesis addresses.

3. THEORETICAL BACKGROUND AND SYSTEM DESIGN

This chapter establishes the theory that motivates the design of the system and then describes the system itself. Sections 3.1 to 3.3 introduce forecast-combination theory, map it onto the trading problem, and state the three hypotheses that the experiments test. Sections 3.4 to 3.7 describe the architecture, the five signal agents, the decision agent that aggregates them, and the taxonomy that makes the heterogeneity of the agent pool explicit.

Throughout the thesis, the term *ensemble* denotes the complete five-agent system together with its decision agent, as opposed to the single-agent and leave-one-agent-out variants used as baselines in Chapter 5.

3.1. Forecast combination and variance reduction

Forecast-combination theory supplies the reasoning behind the decision agent. Consider K forecasts $f_1, \dots, f_K$ of a target y that is not observable at the time of forecasting. Write $e_k = y - f_k$ for the error of the k th forecast, σ_k^2 for the variance of that error and ρ_{kj} for the correlation between the errors of forecasts k and j . Any convex combination, meaning any set of weights $w_k \geq 0$ summing to one, then carries a combined error whose variance is

$$\text{Var}\left(y - \sum_{k=1}^K w_k f_k\right) = \sum_{k=1}^K \sum_{j=1}^K w_k w_j \rho_{kj} \sigma_k \sigma_j \quad (3.1)$$

and this is generally smaller than the variance of the individual forecasts entering it, provided they are not perfectly correlated [23, 24, 25]. Empirical work over several decades has found that even the simplest combinations, equal-weight averages among them, tend to beat their individual members across a wide range of settings [24, 25].

The two-forecaster case repays working through, because it exhibits every property the later design relies on. Setting $K = 2$ and placing weight w on f_1 gives a combined error $e_c = w e_1 + (1 - w) e_2$, at which (3.1) reduces to

$$\text{Var}(e_c) = w^2 \sigma_1^2 + (1 - w)^2 \sigma_2^2 + 2w(1 - w) \rho \sigma_1 \sigma_2. \quad (3.2)$$

Differentiating with respect to w and equating the derivative to zero yields the variance-minimising weight

$$w^* = \frac{\sigma_2^2 - \rho \sigma_1 \sigma_2}{\sigma_1^2 + \sigma_2^2 - 2\rho \sigma_1 \sigma_2}, \quad (3.3)$$

which is the classical result of [23]. Substituting w^* back into (3.2) gives the smallest variance the pair can attain,

$$\text{Var}(e_c^*) = \frac{\sigma_1^2 \sigma_2^2 (1 - \rho^2)}{\sigma_1^2 + \sigma_2^2 - 2\rho \sigma_1 \sigma_2}. \quad (3.4)$$

Three readings of (3.2) to (3.4) shape everything that follows. Putting $\rho = 0$ into (3.3) leaves $w^* = \sigma_2^2 / (\sigma_1^2 + \sigma_2^2)$, so that under uncorrelated errors the optimal weights are proportional to inverse variances, $w_k \propto 1/\sigma_k^2$. That is the precision-weighting intuition Section 3.2 borrows and Section 5.2.8 eventually puts to the test. Putting $\sigma_1 = \sigma_2 = \sigma$ and $w = 1/2$ instead reduces (3.2) to $\sigma^2(1 + \rho)/2$, which sits below σ^2 for every $\rho < 1$ and keeps falling as ρ does; equally accurate forecasters need only to disagree, not to be individually strong. And

since $\partial \text{Var}(e_c)/\partial \rho = 2w(1-w)\sigma_1\sigma_2 > 0$ at any interior weight, the combined variance rises monotonically with correlation, so a negatively correlated pair does strictly better than an uncorrelated one. Deliberately opposed agents are therefore an asset to be sought rather than an awkwardness to be tolerated.

Two consequences of the general form (3.1) bear directly on the design of a multi-agent trading system. The first has just been derived in miniature: because the combined variance rises with every ρ_{kj} , agents that disagree are worth more than agents that agree, and agents that disagree systematically are worth most of all. That is the argument for populating the pool with methodologically complementary members rather than with variations on a single idea, and it predicts, for instance, that a momentum-family agent and a mean-reversion-family agent will produce negatively correlated signals whenever the market is trendless.

The second is that the mechanism degrades gracefully. Nothing in (3.1) privileges any particular member of the pool, so the variance reduction persists for as long as two imperfectly correlated forecasters remain within it. Section 5.3.1 turns that prediction into the leave-one-agent-out ablation.

3.2. Mapping to the trading problem

Carrying that framework over to trading requires one identification. On date t and for symbol s , every signal agent a emits a direction $\sigma_{a,s}(t) \in \{-1, 0, +1\}$ paired with a confidence $c_{a,s}(t) \in [0, 1]$, and it is the confidence that is made to stand in for the inverse variance appearing in (3.3). The stand-in is heuristic, monotone and bounded: a larger value means the indicator of the agent has moved further inside its active region, which the design reads as a sign that the agent is on firmer ground at that moment.

Those per-agent readings are aggregated by the decision agent into a single score

$$\text{score}(t, s) = \sum_a w_{\kappa(a)} c_{a,s}(t) \sigma_{a,s}(t), \quad (3.5)$$

in which $\kappa(a) \in \{\text{trend, momentum, mean-reversion, volatility}\}$ names the methodological family that agent a belongs to, and $w_{\kappa(a)}$ is a weight attached to that family, serving as a prior on how far its members are to be trusted relative to the others. Section 3.6 fixes both those weights and the thresholds that turn the score, symbol by symbol, into a buy, a sell or a hold.

The standing of this identification is limited in two ways that should be carried through the rest of the thesis. The first is that signals and confidences alike are heuristic transformations of price history rather than calibrated statistical forecasts. Predictive distributions are estimated nowhere in the system, and no confidence value is fitted to data of any kind. What connects the design to classical forecast-combination theory is accordingly an analogy and not a derivation: (3.1) argues for an architecture of methodologically heterogeneous agents whose errors are imperfectly correlated, and it argues for giving the score (3.5) its linear form, but nothing in it establishes that $c_{a,s}(t)$ behaves like an estimate of $1/\sigma_a^2$.

The second is that the analogy is testable, and that it is duly tested. Whether a higher confidence really does mark a more accurate signal is settled in Section 5.2.8, which relates confidence values to realised directional accuracy over the training window. The hypotheses stated next are framed so as not to rest on the answer: they concern the risk-adjusted behaviour of the pooled ensemble, which (3.1) predicts from error diversity by itself, with no appeal to precision weighting.

3.3. Hypotheses

Three testable hypotheses follow from that mapping. The first casts the ensemble as a *risk-aware* decision support tool, in keeping with what forecast-combination theory actually predicts, which is a reduction in variance rather than any systematic lift in the mean.

- **H1.** Measured on held-out data with transaction costs deducted, the multi-agent system matches or betters buy-and-hold WIG20 in three respects: across the risk-adjusted measures (Sharpe, Sortino, Calmar, Ulcer, Martin), in maximum drawdown, and in a comparison of equity curves matched on volatility. This holds even should the system fail to dominate on gross total return.
- **H2.** Measured on held-out data with transaction costs deducted, the multi-agent system beats the strongest single-agent baseline in total return. The reasoning is that a combined score spreads itself over agents whose errors differ, and should therefore surpass any one of them, barring the degenerate case where a single agent is uniformly the best of the pool.
- **H3.** The performance of the ensemble withstands the removal of any single signal agent. The reasoning is that no one agent carries the directional signal by itself, and that variance reduction survives, by (3.1), for as long as two imperfectly correlated forecasters are left in the pool.

WIG20 being the primary panel, H1 is posed against buy-and-hold on that index, while Section 5.3.7 carries the identical battery of risk-adjusted comparisons over to the FTSE 100 by way of supplementary external evidence. H2 and H3 are tested on WIG20 and nowhere else. Chapter 5 presents the evidence bearing on each and weighs how far it agrees with what the theory predicts.

3.4. System architecture

Four layers make up the system, as Fig. 3.1 shows. At the bottom the *data layer* reads daily closing prices for the WIG20 constituents into one common panel format, disposing of missing values by dropping whichever rows are incomplete. Above it the *signal layer* holds the five independent signal agents, each of which receives the history of one symbol, truncated at the current date inclusive, and returns a triple: a direction drawn from $\{-1, 0, +1\}$, a confidence drawn from $[0, 1]$, and the name of a methodological family, whether trend, momentum, mean-reversion or volatility.

The *decision layer* above that consists of one decision agent, which collects every per-symbol signal issued on a given date and returns a single recommendation per symbol, either buy, sell or hold. The *presentation layer* at the top holds two components. A portfolio simulator executes the recommendations at the daily closing price, keeps cash and positions after commissions and slippage, and records the resulting equity curve. A metrics calculator then derives from that curve the risk-adjusted metrics reported in Section 5.1.6.

A common base class, from which every signal agent inherits, guarantees the uniformity of that interface. Each agent is confined in addition to one family of indicators, with no mixing of rules, and that restriction is what keeps the aggregation argument of Section 3.1 well defined, since it allows each agent to stand for a distinct forecaster f_k carrying its own bias-variance profile. Chapter 4 describes how the four layers are realised in software.

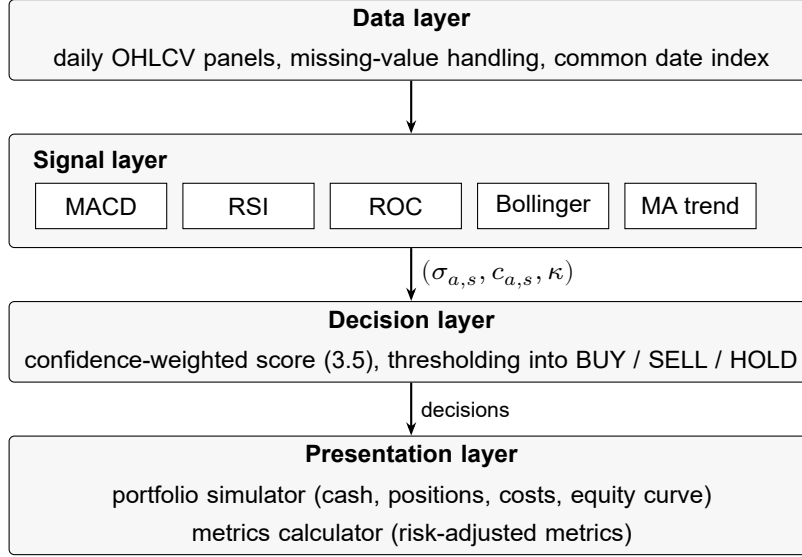


Fig. 3.1. System architecture. The four layers communicate through well-defined interfaces, so that new signal agents can be added without changing the decision layer.

3.5. Signal agents

Every signal agent plays the part of an independent and imperfect forecaster of the next directional move in a single symbol. Confining each one to a single methodological family is what makes the forecast-combination argument of Section 3.1 well defined, and it is also what drives the forecast errors of any two agents as far apart as the indicator set permits.

The indicators themselves are not contributions of this thesis, and the boundary between what is borrowed and what is original deserves to be drawn explicitly. Three of the five come from identifiable authors, and this work adopts not only their definitions but the conventional parameter settings that accompany them, which is what licenses the claim, made here and repeated in Section 5.1.3, that no indicator hyperparameter is ever tuned. Provenance is accordingly stated agent by agent below. What is original to this work is the layer constructed on top of the indicators: the confidence function attached to each one, the assignment of agents to methodological families, and the family weights of (3.10). None of the three has a source in the technical-analysis literature, and each is presented here as a design decision of the author rather than as received practice.

Each entry below gives the computation the agent performs, the source of the indicator it computes, and the forecasting role it fills, meaning the market behaviour it is built to detect together with the circumstances under which it tends to be wrong. Two conventions hold throughout: an agent returns a neutral hold signal whenever the available history is too short for its indicator, and confidence is zero everywhere outside the active signal band.

MACD (moving-average convergence-divergence). The indicator is due to Appel [45]. Subtracting a slow exponential moving average (EMA) of the closing price from a fast one, at default spans of 17 and 8 respectively, produces the MACD line, and taking an EMA of that line in turn, at a default span of 9, produces the signal line. An upward crossing of the two lines makes the agent emit +1, a downward crossing −1. The confidence attached,

$$\text{confidence} = \tanh\left(\frac{|h|}{2\sigma}\right), \quad (3.6)$$

is the author's own and has no counterpart in [45]; here h is the current histogram value and σ its standard deviation over the preceding 50 bars, so that the hyperbolic tangent carries a normalised

histogram magnitude into $[0, 1]$. Role: a momentum forecaster reporting how far recent directional moves have persisted, whose errors gather at trend reversals, where a crossover necessarily lags the turning point. Methodological family: momentum.

RSI (relative strength index). The index was introduced by Wilder [46]. It compares the average gain with the average loss over a rolling window and is defined as $RSI = 100 - 100/(1 + RS)$, where $RS = \text{avg_gain}/\text{avg_loss}$. The action rule is

$$\text{action} = \begin{cases} \text{buy,} & RSI < 30, \\ \text{sell,} & RSI > 70, \\ \text{hold,} & \text{otherwise,} \end{cases} \quad (3.7)$$

in which the window of 14 days and the thresholds of 30 and 70 are Wilder's own settings, taken over here unaltered. Confidence is the author's addition and equals the normalised distance of the index from whichever threshold lies nearer. Role: a mean-reversion forecaster betting against recent price extremes, whose errors are largest in strong trends, when the index can sit overbought or oversold for extended stretches. Methodological family: mean-reversion.

ROC (rate of change). Unlike the two preceding indicators the rate of change has no single originator, being standard material in the reference literature on technical indicators [47, 48]. It is simply the percentage change in the closing price across a fixed number of days, here 10. The action rule is

$$\text{action} = \begin{cases} \text{buy,} & ROC > \tau, \\ \text{sell,} & ROC < -\tau, \\ \text{hold,} & \text{otherwise,} \end{cases} \quad (3.8)$$

where the threshold τ , set by default to 0.02, and the confidence, taken as $|ROC|$ clipped to $[0, 1]$, are both choices of the author rather than conventions of the literature. Role: a momentum forecaster working on a shorter horizon than MACD and responding to the bare speed of price change, whose errors arise wherever a fast move reverses without warning. Methodological family: momentum.

Bollinger Bands. The construction is Bollinger's [49]. A simple moving average (SMA) of the closing price over a window supplies the middle band, and the outer bands are placed at $U = \text{middle} + m\sigma$ and $L = \text{middle} - m\sigma$. The action rule is

$$\text{action} = \begin{cases} \text{buy,} & \text{close} < L, \\ \text{sell,} & \text{close} > U, \\ \text{hold,} & \text{otherwise.} \end{cases} \quad (3.9)$$

The window of 20 days and the multiplier $m = 2$ employed here are Bollinger's own specification. Confidence, again supplied by the author, is the normalised distance of the close from the nearer band. Role: a volatility forecaster gauging, in units of the standard deviation, how far price has pulled away from its mean, and calling a reversion once the close pierces a band; its errors occur where a breakout through a band, rather than reverting, matures into a sustained move. Since the bands respond to dispersion and not to direction, this agent carries less directional information than any of the other four, which its family weight duly reflects (Section 3.6). Methodological family: volatility.

MA trend. Two simple moving averages are compared, a fast one over 50 days and a slow one over 200. The pairing is long-standing practice in the technical-analysis literature [48],

and the predictive content of moving-average crossover rules on equity data is the subject of [2]. A buy signal is emitted when the fast SMA exceeds the slow SMA, and a sell signal in the opposite case. Confidence, set by the author, equals the absolute difference between the two averages divided by the latest close. Role: a long-horizon trend forecaster reading the prevailing market regime off the relation between a short and a long average, whose errors accumulate in choppy, trendless markets where the two averages cross one another repeatedly. Methodological family: trend.

3.6. Decision agent

Signals produced on a given date arrive at the decision agent as a list, which it groups by symbol. Any signal whose confidence falls below a global minimum $c_{\min}$ is discarded outright, and whatever survives is aggregated into the score defined in (3.5). The family weights w_{κ} entering that score are held fixed at

$$\begin{aligned} w_{\text{trend}} &= 1.3, & w_{\text{momentum}} &= 1.0, \\ w_{\text{mean-rev.}} &= 0.8, & w_{\text{volatility}} &= 0.5. \end{aligned} \tag{3.10}$$

Encoded in these four numbers is a fixed prior on how far each methodological family may be trusted to call direction, and the ordering derives from the empirical literature rather than from any in-sample search. The largest weight goes to trend-following, because time-series momentum stands among the best-documented return predictors there are, robust and persistent alike across a range of equity markets and holding horizons [33, 32], and because daily index constituents of the WIG20 sort tend to exhibit sustained directional regimes. Shorter-horizon momentum takes a unit weight and serves as the reference family. Mean reversion sits below momentum on the grounds that overbought and oversold rules inform mainly in range-bound phases and are frequently wrong through strong trends. Volatility takes the smallest weight of the four, since band-based signals answer to dispersion rather than to direction and so carry the least directional information.

The four levels are deliberately coarse, equally spaced multiples chosen to keep the weights interpretable and the degrees of freedom low. They stay fixed across every symbol, every regime and every evaluation window, and they are never tuned on test data.

Declining to estimate the weights at all may look like a concession, but the forecast-combination literature suggests it is closer to a precaution. Section 3.1 derived the variance-minimising weight (3.3) on the assumption that σ_1 , σ_2 and ρ are known quantities. In application they are not. Each must itself be estimated from a finite sample, which makes the resulting $\hat{w}$ a random variable whose sampling error feeds straight into the variance of the combination. Where the pooled forecasters are of comparable accuracy the surface (3.2) is nearly flat about its optimum, so the theoretical gain from departing from equal weights is small while the estimation error incurred in hunting for that optimum is not. The two effects routinely cancel, and the second frequently dominates, with the consequence that simple fixed weights prove obstinately difficult to beat with estimated ones. The phenomenon is established enough to have earned a name, the *forecast combination puzzle* [50, 51], and it is why a fixed prior is preferred here to a fitted weight vector.

How far the results depend on that scheme is settled empirically in Section 5.3.2, which perturbs each family weight by up to $\pm 50\%$ and then replaces the vector wholesale with equal, inverted, in-sample-optimised and inverse-variance-estimated alternatives. The limiting case, removing an agent altogether, is covered by the leave-one-agent-out ablation of Section 5.3.1, and regime-conditional weighting is taken up as future work in Section 6.2.

Comparing the score against mode-dependent thresholds then yields the final decision:

$$\text{action} = \begin{cases} \text{BUY,} & \text{score} \geq \theta_{\text{buy}}, \\ \text{SELL,} & \text{score} \leq \theta_{\text{sell}}, \\ \text{HOLD,} & \text{otherwise.} \end{cases} \quad (3.11)$$

Those thresholds are $\theta_{\text{buy}} = -\theta_{\text{sell}} = 1.2$ under *passive* mode, 0.7 under *balanced* mode and 0.3 under *aggressive* mode. Passive mode imposes one further condition, namely that the sign of the score agree with the dominant trend direction, itself defined as the sign-weighted sum of the confidences of the trend-family agents.

The decision agent is governed by two parameters: the mode m , which takes one of the values *passive*, *balanced* or *aggressive*, and the minimum confidence threshold c_{min} . These are the only quantities anywhere in the system that walk-forward evaluation is permitted to tune. Every indicator hyperparameter remains at the literature-conventional default recorded in Section 3.5 throughout.

3.7. Agent taxonomy

Three dimensions summarise the five signal agents in Table 3.1: the methodological family each belongs to, the type of signal it produces, whether by crossover or by threshold, and the horizon of the indicator beneath it. Section 3.5 supplies the forecasting rationale of each family alongside the agent descriptions.

Setting the agents out this way makes the heterogeneity of the pool explicit and motivates the family-level weights w_{κ} , since agents expected to make the most distinct forecast errors are pooled across families carrying different weights, exactly as the variance-reduction argument of Section 3.1 recommends. The measured correlations bear the taxonomy out. Section 5.2.7 locates the largest absolute correlations in cross-family pairs, RSI against ROC and Bollinger against ROC among them, and finds them carrying opposing signs, which is what two rules reading the same raw price data in opposite ways ought to produce.

Table. 3.1. Taxonomy of the five signal agents. Horizon refers to the nominal lookback of the underlying indicator.

Agent	Family	Signal type	Horizon
MACD	momentum	crossover	17 d
RSI	mean-reversion	threshold	14 d
ROC	momentum	threshold	10 d
Bollinger	volatility	threshold	20 d
MA trend	trend	crossover	200 d

3.8. Chapter summary

Forecast-combination theory predicts that pooling imperfectly correlated forecasters reduces the variance of the combined forecast, and that the reduction is largest when the pooled forecasters make errors of different kinds. The system described in this chapter is built to instantiate that prediction: five agents, each confined to a single methodological family, feed a decision agent that combines their signals into the linear, confidence-weighted score of (3.5) and thresholds it into a

trading action.

Two design choices deserve emphasis, because the evaluation returns to both. The family weights are a fixed prior taken from the empirical literature rather than an estimated quantity, which trades potential in-sample fit for robustness. The confidence values are heuristics rather than calibrated probabilities, which means the inverse-variance reading of the score is an analogy to be tested, not an established property. Chapter 5 tests both, along with the three hypotheses stated in Section 3.3.

4. IMPLEMENTATION

Chapter 3 described the system as a set of interacting abstractions: agents that emit signals, a decision agent that combines them, a simulator that executes the resulting orders. This chapter describes the software that realises those abstractions. It states the requirements the implementation had to satisfy, justifies the selected technologies, presents the structure of the code in UML form, walks through the execution of a single trading day, describes the preparation of the price data, catalogues the experiment harness that produces the results of Chapter 5, and explains how the implementation was verified.

The complete system comprises 46 Python modules, of which 16 are experiment scripts.

4.1. Requirements

The requirements were derived from the aim stated in Section 1.2 and from the evaluation protocol the system had to support. They are stated here in the form in which they governed the implementation.

4.1.1. Functional requirements

Table 4.1 lists the functional requirements. Requirements F1 to F4 follow from the system design of Chapter 3; F5 to F8 follow from the evaluation protocol, which needs to run the same pipeline hundreds of times under varying configurations.

Table. 4.1. Functional requirements of the system.

ID	Requirement
F1	Load daily OHLCV data for a configurable roster of symbols from local CSV files.
F2	Compute, for each agent, symbol and date, a directional signal, a confidence value and a methodological family, using only data available up to that date.
F3	Aggregate all signals for a symbol on a date into a single buy, sell or hold decision through the confidence-weighted score of (3.5).
F4	Simulate the execution of decisions on a cash portfolio, charging commission, slippage and, where applicable, a one-way transaction tax.
F5	Compute the risk-adjusted metrics of Section 5.1.6 from the resulting equity curve and trade list.
F6	Run an anchored walk-forward search over decision-agent parameters and evaluate the selected configuration on a held-out window.
F7	Run ablation, weight-sensitivity, cost-sensitivity, regime, bootstrap and confidence-validation experiments over the same pipeline.
F8	Export every experiment result to CSV so that the numbers reported in the thesis are traceable to a file.

4.1.2. Non-functional requirements and design constraints

The non-functional requirements are listed in Table 4.2. Two of them deserve comment, because they shaped the code more than the others.

Requirement N1, the absence of look-ahead, is a correctness property rather than a preference: a single leak invalidates the entire evaluation. It is enforced structurally rather than by

convention. Each agent receives a data frame already truncated at the current date, so an agent cannot observe a future bar even if its implementation were careless.

Requirement N3, the separation of signal computation from portfolio accounting, began as a performance concern and became an architectural principle. The experiment battery re-runs the backtest across grid points, ablations, weight perturbations and cost configurations, and recomputing indicator values on every run made the full battery impractically slow. Precomputing the signal table once and passing it into each backtest reduced the end-to-end runtime by more than an order of magnitude.

Table. 4.2. Non-functional requirements and design constraints.

ID	Requirement
N1	No look-ahead: a signal for date t must depend only on data up to and including t .
N2	Determinism: repeated runs on identical inputs must produce identical outputs, so that reported numbers are reproducible.
N3	Separation of signal computation from portfolio accounting, so that cost and parameter sweeps do not recompute indicators.
N4	Extensibility: a new signal agent must be addable without modifying the decision layer or the simulator.
N5	Configurability: symbols, data paths, costs, decision parameters and agent hyperparameters must be declared outside the code.
N6	Market portability: switching to another exchange must require only a new data source and cost calibration.
C1	Every signal agent inherits from BaseAgent and implements a single method.
C2	Every signal agent uses one internally consistent methodology; mixed-indicator agents are not permitted.
C3	Signal agents may not make portfolio-level decisions; they only emit signals.
C4	Every signal carries a confidence value alongside its direction.

Constraints C1 to C4 are architectural rules adopted at the start of the project. C2 is the one with theoretical weight: it is what makes the forecast-combination argument of Section 3.1 well defined, because an agent mixing several indicator families would not correspond to a single forecaster with an identifiable error profile.

4.2. Technology stack

The system is implemented in Python 3.11. Python was selected because the numerical and machine-learning libraries required by the evaluation are mature and interoperable in that ecosystem, and because the resulting code is readable enough to serve as documentation of the method, which matters for a thesis whose contribution is partly methodological.

Table 4.3 lists the libraries and the role of each. The dependency set is deliberately small: five libraries carry the entire multi-agent system, and the remaining two are needed only for the machine-learning baselines. That asymmetry is itself a result. The ensemble requires no training infrastructure at all, which is the deployment argument made in Section 5.4.7.

Two choices merit justification. Rolling-window indicator computation is delegated to pandas rather than implemented manually, because the library implementations are well tested and their treatment of leading undefined values is explicit, which matters for N1. LightGBM was chosen for the second machine-learning baseline in preference to another neural architecture, because a

Table. 4.3. Technologies used in the implementation, with the role of each.

Component	Version	Role
Python	3.11	Implementation language.
pandas	≥ 2.0	Price panels, rolling-window indicator computation, equity curves.
NumPy	≥ 1.24	Vectorised numerics underlying the metric calculations.
Matplotlib	≥ 3.7	Generation of the figures reported in Chapter 5.
PyYAML	≥ 6.0	Declarative configuration of symbols, costs, agents and decision parameters.
SciPy	≥ 1.10	Spearman rank correlation used in the confidence validation of Section 5.2.8.
PyTorch	≥ 2.2	LSTM directional baseline of Section 5.1.5.
LightGBM	≥ 4.0	Gradient-boosted-trees baseline of Section 5.1.5.

gradient-boosted tree ensemble is structurally different from a recurrent network, so agreement between the two baselines is more informative than agreement between two variants of the same model family.

4.3. Software architecture

The four conceptual layers of Section 3.4 map onto Python packages as follows. The data layer is a loader function operating on a directory of CSV files. The signal layer is the package `src/agents`, containing the five agents and their common base. The decision layer is a single class, `DecisionAgent`. The presentation layer is the package `src/utils`, containing the portfolio simulator, the metrics calculator, and the plotting helpers. The experiment harness in `scripts/sits` above all four.

Fig. 4.1 shows the class structure of the signal and decision layers.

4.3.1. The agent contract

The base class is deliberately minimal. It declares two class attributes, `name` and `kind`, the latter being the methodological family $\kappa(a)$ of Section 3.2; it provides two helpers that construct the return dictionary; and it declares a single abstract method that every agent must implement. Listing 4.1 reproduces it in full.

Three properties of this contract are worth noting. The `_pack` helper clips confidence into $[0, 1]$, so no agent can inflate its own weight in the aggregate score by returning an out-of-range value. The `_neutral_signal` helper gives every agent a uniform way to abstain, which agents use when the price history is too short to compute their indicator. And because the only input is a data frame and a symbol, the interface says nothing about portfolio state, which enforces constraint C3 structurally.

Adding a new agent therefore means writing one class with one method, and registering it in the configuration file. Neither the decision agent nor the simulator changes, which satisfies N4.

4.3.2. The decision layer

`DecisionAgent` implements (3.5) and the thresholding rule (3.11) directly. Its `decide` method groups the signals of a date by symbol, delegates to a per-symbol routine, discards hold decisions, and applies an optional cap on the number of simultaneous positions. Listing 4.2 shows the per-symbol routine, which is the concrete realisation of the score.

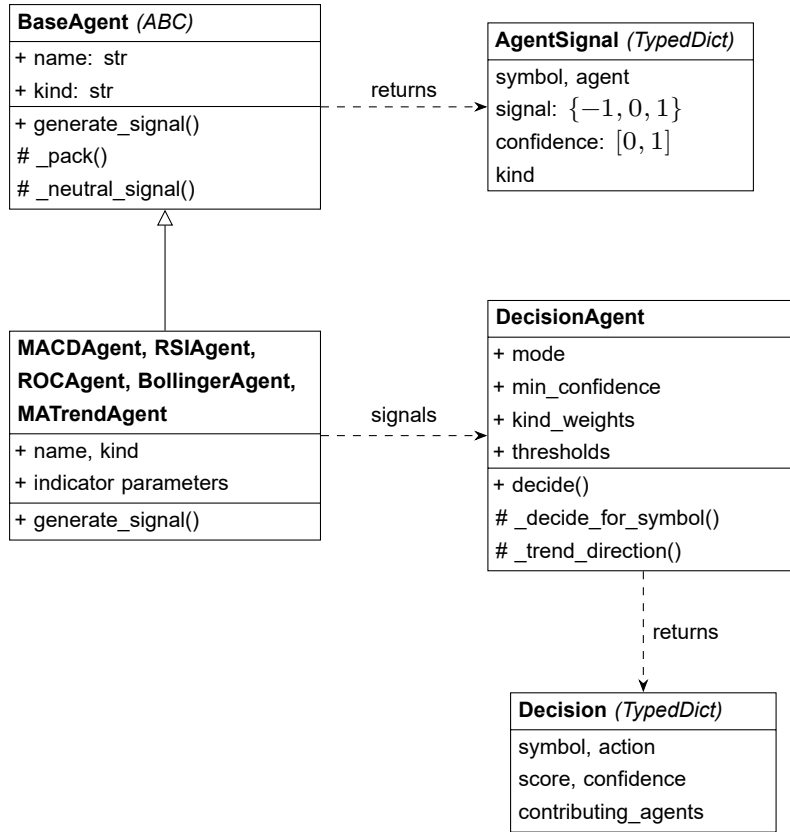


Fig. 4.1. Class structure of the signal and decision layers. All five signal agents inherit from the abstract base class and communicate only through the AgentSignal dictionary, so the decision layer is independent of how any individual signal is computed.

The correspondence with the theory is close enough to read line by line. The filter on `min_confidence` is the threshold $c_{\min}$; the product of signal, confidence and family weight is one term of the sum in (3.5); and the comparison against the mode-dependent thresholds is (3.11). The additional branch under *passive* mode implements the trend-agreement requirement described in Section 3.6.

One implementation detail carries analytical value. Every decision records the list of contributing agents together with their individual signals and confidences. This is what makes the per-agent attribution described in Section 5.4.6 possible: the score of any day can be decomposed after the fact into the contribution of each agent, without re-running the pipeline.

4.4. Implementation of the trading pipeline

Fig. 4.2 shows the execution of a single trading day. The loop over dates is driven by the backtest function, which holds the precomputed signal table and calls the decision agent and the simulator in turn.

4.4.1. Signal precomputation and the absence of look-ahead

The signal table is built once by iterating over the dates common to all symbols and, for each date and symbol, calling every agent with the price history truncated at that date. The truncation is the mechanism that enforces N1: an agent is handed `df.loc[:date]` and therefore cannot observe a future bar.

This construction is safe even though it uses the full price history, because the signal for

```

1  from abc import ABC, abstractmethod
2  import pandas as pd
3  from src.agents.base.AgentSignal import AgentSignal
4
5
6  class BaseAgent(ABC):
7      name: str
8      kind: str
9
10     def _neutral_signal(self, symbol: str) -> AgentSignal:
11         return {
12             "symbol": symbol,
13             "agent": self.name,
14             "signal": 0,
15             "confidence": 0,
16             "kind": self.kind
17         }
18
19     def _pack(self, symbol: str, signal: int, confidence: float) -> AgentSignal:
20         return {
21             "symbol": symbol,
22             "agent": self.name,
23             "signal": signal,
24             "confidence": min(max(confidence, 0.0), 1.0),
25             "kind": self.kind
26         }
27
28     @abstractmethod
29     def generate_signal(self, data: pd.DataFrame, symbol: str) -> AgentSignal:
30         raise NotImplementedError

```

Listing 4.1. The abstract base class defining the agent contract.

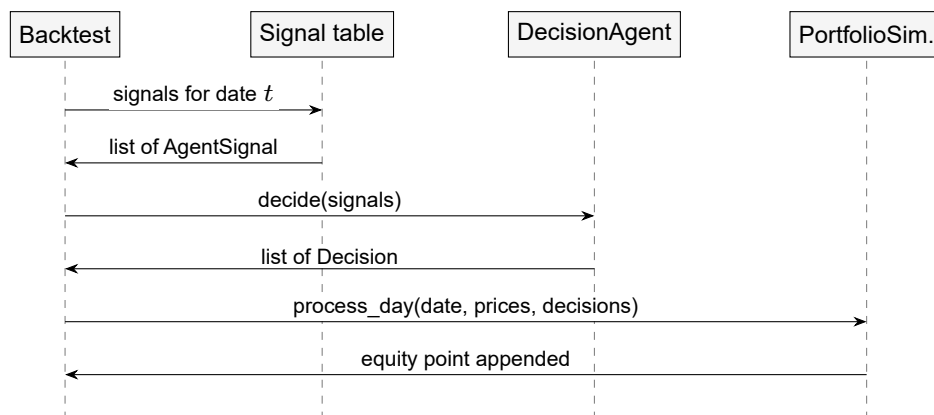


Fig. 4.2. Execution of one trading day. Signals are read from the precomputed table rather than recomputed, which is what makes the parameter and cost sweeps of Chapter 5 affordable.

```

1  def _decide_for_symbol(self, symbol: str, signals: List[dict]) -> Decision:
2      filtered = [
3          s for s in signals
4          if s["signal"] != 0 and s["confidence"] >= self.min_confidence
5      ]
6
7      if not filtered:
8          return self._hold(symbol)
9
10     trend_dir = self._trend_direction(signals)
11     score = 0.0
12     contributors = []
13
14     for s in filtered:
15         if self.mode == "passive" and trend_dir != 0:
16             if s["signal"] != trend_dir:
17                 continue
18
19         weight = self.kind_weights.get(s["kind"], 1.0)
20         contribution = s["signal"] * s["confidence"] * weight
21         score += contribution
22         contributors.append({
23             "agent": s["agent"],
24             "signal": s["signal"],
25             "confidence": s["confidence"],
26             "kind": s["kind"],
27         })
28
29     thresholds = self.thresholds[self.mode]
30
31     if score >= thresholds["buy"]:
32         return self._decision(symbol, "BUY", score, contributors)
33     elif score <= thresholds["sell"]:
34         return self._decision(symbol, "SELL", score, contributors)
35     else:
36         return self._hold(symbol)

```

Listing 4.2. Per-symbol aggregation: the implementation of the confidence-weighted score of (3.5) and of the thresholding rule of (3.11).

date t depends only on data up to t by construction. Querying the resulting table over a later sub-window therefore introduces no leakage. That property is what allows the same table to serve the training window, the test window, every ablation and every cost configuration.

4.4.2. Portfolio accounting

The simulator maintains cash, a dictionary of open positions, a trade log and an equity curve. On each day it processes sell decisions before buy decisions, so that capital released by a closing position is available to a new one on the same date, and then records the mark-to-market value of the portfolio.

The cost model of Section 5.1.2 is implemented in the fill computation. A purchase fills at $p(1 + s)$ and a sale at $p(1 - s)$, where s is the one-way slippage rate, so the round trip pays slippage twice. Commission is charged on both legs. A one-way transaction tax, used for the UK Stamp Duty Reserve Tax in Section 5.3.7, is charged on purchases only and is kept as a separate

rate so that the two legs can carry asymmetric charges.

Position sizing accounts for entry charges rather than ignoring them. The allocation for a new position is a fixed fraction of current cash, and the affordable notional is that allocation divided by $1 + \gamma + \tau$, where γ is the commission rate and τ the buy-side tax rate. The share count is then the integer floor of the affordable notional divided by the fill price, so the simulator never buys fractional shares and never spends cash it does not hold. Listing 4.3 shows the routine.

```
1
2     # Size the position against cash capacity inclusive of entry charges
3     allocation = self.cash * self.position_size
4     max_spend = allocation / (1.0 + buy_charge_rate)
5     if max_spend <= 0:
6         return
7
8     quantity = int(max_spend // fill_price)
9     if quantity <= 0:
10        return
11
12    notional = quantity * fill_price
13    entry_charge = notional * buy_charge_rate
14    total_cost = notional + entry_charge
15    if total_cost > self.cash:
16        return
17
18    self.cash -= total_cost
19
20    self.positions[symbol] = {
21        "symbol": symbol,
22        "quantity": quantity,
23        "entry_price": fill_price,
24        "entry_date": date,
25        "entry_commission": entry_charge,
26    }
27
28    self.trades.append({
29        "symbol": symbol,
30        "action": "BUY",
31        "price": fill_price,
32        "quantity": quantity,
33        "date": date,
34        "pnl": None,
35    })
36
37    def _sell(self, symbol: str, price: Optional[float], date: pd.Timestamp) -> None:
38        if price is None:
39            return
40        if symbol not in self.positions:
41            return
```

Listing 4.3. Entry-side accounting in the portfolio simulator, including slippage, commission and the one-way buy tax.

Realised profit on a closed trade is computed net of both commissions and of the round-trip slippage already embedded in the two fill prices, which is why the entry commission is stored with the position rather than discarded after the purchase.

4.4.3. Metrics

`MetricsCalculator` consumes the equity curve and the trade log and implements the definitions of Section 5.1.6. Annualisation uses $\sqrt{252}$ for ratio statistics and calendar years, computed as elapsed days divided by 365.25, for CAGR. Each metric guards against the degenerate inputs that arise during the parameter sweep, in particular an empty trade list and a flat equity curve, and returns zero rather than raising, so that a configuration that never trades is comparable rather than fatal. This is the behaviour that makes the *passive* column of Table 5.1 well defined.

4.5. Data preparation

Price data are daily OHLCV series downloaded from Stooq [52] as one CSV file per instrument, with columns Date, Open, High, Low, Close and Volume. Files are named by the lower-case ticker, and the loader skips any symbol declared in the configuration that has no corresponding file, so a partial data directory degrades gracefully rather than failing.

Three preparation steps are applied. First, all series are truncated at the common end date of 2026/03/31, so that no instrument contributes observations the others cannot. Second, rows with missing values are dropped per symbol. Third, the backtest intersects the date indices of all loaded symbols and iterates only over dates common to every instrument, which guarantees that the portfolio sees a consistent cross-section on every date.

The WIG20 roster comprises nineteen constituents and the index series. Žabka Group SA. is excluded because its listing on 2024/10/17 falls inside the test window, so including it would introduce a constituent whose history is absent from the training window. The FTSE 100 panel is prepared identically, with 20 constituents and the index series, and is held in a separate directory selected through the configuration file.

4.6. Experiment harness

Each experiment reported in Chapter 5 is produced by a dedicated script that loads the configuration, builds the agents, runs the relevant sweep and writes a CSV file. Table 4.4 maps the experiments to their scripts and outputs.

All runtime parameters live in a YAML file rather than in code, which satisfies N5. The configuration declares the data directory, the symbol roster, initial cash and position size, the commission and slippage rates, the anchored split date and fold count, the decision-agent mode and minimum confidence, the enabled agents with their indicator parameters, and the hyperparameter grids of the two machine-learning baselines. Disabling an agent is a matter of removing one line.

Market portability, requirement N6, is realised through this file. The FTSE 100 replication of Section 5.3.7 uses a second configuration that differs from the primary one only in the data directory, the symbol roster and the cost calibration. No code path is specific to either market.

4.7. Verification of the implementation

Because the contribution of this work rests on the credibility of its numbers, verification was directed at the properties that would invalidate them if violated, rather than at line coverage.

Absence of look-ahead. The critical property, N1, is verified structurally rather than by test. Agents receive a truncated data frame and have no access to the full panel, so a leak would require an agent to reach outside its argument. The precomputation step preserves this property because a signal at date t is a function of data up to t only, which makes the table equivalent to computing

Table. 4.4. Experiment harness: the script that produces each result reported in Chapter 5, and the file it writes.

Experiment	Script	Output
Walk-forward selection and headline metrics	run_walk_forward.py	walk_forward_*.csv
Leave-one-agent-out ablation	run_ablation.py	ablation.csv
Cost-sensitivity sweep	run_cost_sensitivity.py	cost_sensitivity.csv
Risk-adjusted and volatility-matched metrics	run_risk_metrics.py	risk_metrics*.csv
Market-regime decomposition	run_regime_analysis.py	regime_analysis.csv
Bootstrap intervals and paired tests	run_stats.py	stats.csv
Weight sensitivity	run_weight_sensitivity.py	weight_sensitivity.csv
Confidence validation	run_confidence_validation.py	confidence_validation_*.csv
Selection-objective comparison	run_objective_comparison.py	objective_comparison.csv
LSTM baseline	run_ml_baselines.py	ml_baseline*.csv
GBM baseline	run_gbm_baseline.py	gbm_baseline*.csv
Figures	regenerate_figures.py, regenerate_extra_figures.py	img/*.pdf

signals date by date inside each backtest. This equivalence was checked directly by running a backtest with precomputed signals and one that recomputes signals per date, and confirming that the two produce identical equity curves.

Determinism. Requirement N2 is verified by re-running each experiment script and comparing the regenerated CSV against the committed one. The multi-agent pipeline contains no stochastic component, so agreement is exact. The two machine-learning baselines are seeded explicitly and average three seeded models, so their outputs are reproducible given the same library versions.

Consistency of the accounting. The portfolio simulator was checked against invariants that must hold for any execution path: cash never becomes negative, because a purchase is rejected when its total cost exceeds available cash; the share count is integral; a position is never opened twice for the same symbol; and the realised profit reported for a closed trade equals the difference in fill prices times the quantity, less both commissions.

Cross-validation of the metrics. The metric implementations were verified against independently computed values on the same equity curves, and against the relationships that must hold between them, in particular that the Calmar ratio equals CAGR divided by the absolute maximum drawdown, and that the Sharpe and Sortino ratios coincide when no daily return is negative.

Verification of the results. The strongest verification of the implementation is the evaluation itself. The battery reported in Chapter 5 exercises the same code paths under hundreds of configurations, and the results move in the directions that the design predicts: removing an agent degrades performance modestly rather than catastrophically, raising costs lowers returns monotonically, and the regime decomposition sums back to the aggregate figures. A defect in the aggregation or accounting would be expected to break at least one of these regularities.

The limitation of this approach should be stated plainly: the implementation carries no automated unit-test suite, so the verification described above establishes the properties that matter for the reported results rather than the correctness of every function in isolation. Adding such a suite is a natural piece of further work.

4.8. Chapter summary

The implementation is small, and deliberately so. Five agents, one decision class, one simulator and one metrics calculator carry the entire system, and the dependency set for that system comprises five widely used libraries with no training infrastructure at all. The complexity that exists lives in the experiment harness rather than in the system under study, which is the appropriate distribution for work whose main claim is about evaluation.

Two implementation decisions had consequences beyond the code. Enforcing one methodological family per agent, constraint C2, is what makes the forecast-combination argument well defined rather than merely suggestive. Separating signal computation from portfolio accounting, requirement N3, is what made the sensitivity analyses of Chapter 5 affordable, and therefore what made the robustness claims possible at all.

5. EVALUATION METHODOLOGY AND RESULTS

This chapter reports the empirical part of the thesis. Section 5.1 describes the evaluation protocol: the datasets, the transaction-cost model, the anchored walk-forward procedure, the baselines, the risk-adjusted metrics, the regime labelling, the volatility-matched comparison, the cost-sensitivity sweep, and the bootstrap tests. Sections 5.2 and 5.3 report the results on the WIG20 panel and the robustness checks together with the external validation on the FTSE 100. Section 5.4 interprets the evidence against the three hypotheses of Section 3.3, and Section 5.5 closes the chapter.

The protocol described here is the element of the work that addresses the four methodological weaknesses identified in Chapter 2, and the stage that the template calls testing and verification of results: every headline number is produced on data that was never used to select a parameter, net of realistic frictions, accompanied by confidence intervals, and stress-tested against ablations, weight perturbations, cost grids, and a second market.

5.1. Experimental setup

5.1.1. Dataset

Daily OHLCV panels for both experiments come from the public repository Stooq [52]. Within a given market the ensemble, the index buy-and-hold benchmark and the machine-learning baselines all trade one fixed roster of constituents, and any incomplete row is dropped from the symbol it belongs to.

The primary panel holds the WIG20 index together with nineteen of its constituent stocks on the Warsaw Stock Exchange. Żabka Group SA. is left out, its listing date of 2024/10/17 being far more recent than that of the remaining constituents. Coverage runs from 2021/05/26 to 2026/03/31, approximately 1,220 trading days, with all values denominated in PLN.

For external validation the panel consists of 20 liquid FTSE 100 constituents plus the FTSE 100 index itself on the London Stock Exchange, spanning the same calendar window and denominated in GBP. Both markets receive an identical anchored split, training up to 2024/06/30 and testing from 2024/07/01 through 2026/03/31. Nothing changes in the agents, the hyperparameters or the evaluation procedures; what differs is confined to the price panels, the index series and the transaction-cost calibration (Sections 5.1.2 and 5.3.7). Section 4.5 describes how both panels were prepared.

5.1.2. Transaction-cost model

Every backtest charges a proportional commission of 39 basis points (bps) of notional on each trade and applies one-way slippage of 5 bps to the executed price, both figures sitting in the range that Polish retail brokerages typically charge on WSE-listed equities. Commission leaves the cash balance on entry and again on exit. Slippage moves the effective buy price up and the effective sell price down by the stated amount. Realised profit or loss is therefore reported after the commissions on both legs and after round-trip slippage.

That calibration errs on the conservative side for a retail investor and cuts substantially into the headline gross return of any high-turnover configuration, which is precisely the methodological concern raised in the literature on backtest reliability [12]. Section 5.2.6 reports a full sensitivity sweep over commission and slippage.

5.1.3. Walk-forward protocol

An anchored walk-forward protocol governs the evaluation. The full period divides into a training window running from 2021/05/26 to 2024/06/30 and a held-out test window running from 2024/07/01 to 2026/03/31. Within the training window a grid search covers the decision-agent parameters

$$\begin{aligned} \text{mode} &\in \{\text{passive}, \text{balanced}, \text{aggressive}\}, \\ c_{\min} &\in \{0, 0.05, 0.10, \dots, 0.30\}, \end{aligned} \tag{5.1}$$

whose in-sample total return proves greatest, and that configuration then goes to the test window untouched. Indicator hyperparameters never enter the grid and stay fixed from beginning to end. Holding the degrees of freedom this low matches the way most of the cited literature treats technical-analysis rules.

Retaining total return as the selection objective is deliberate, notwithstanding that the strategy is judged mainly on risk-adjusted metrics, and the reason is statistical rather than conceptual. Ranking 21 configurations on a low-dimensional grid calls for a simple criterion of low variance, which in-sample total return supplies. Ratio statistics such as the Sharpe ratio do not: on samples of this length they carry substantial estimation error [53], and they degenerate outright for configurations that trade rarely. Since the *passive* mode places no trades at any threshold (Table 5.1), the in-sample Sharpe ratio it yields is undefined rather than simply poor, which would leave any risk-adjusted objective breaking such ties arbitrarily. Section 5.3.6 confirms that nothing rests on the choice, re-running the selection under Sharpe-ratio and Calmar-ratio objectives over the identical grid and finding the selected configuration, and with it every out-of-sample number in this thesis, unchanged on both markets.

A three-fold anchored expanding walk-forward is reported as an additional robustness check. Each fold $k \in \{1, 2, 3\}$ takes as its test slice one of three roughly equal chronological pieces of the held-out window, and trains from 2021/05/26 up to the day preceding that slice. Every fold receives its own fresh grid search, and its winning configuration is evaluated on its own test slice. The point of the exercise is to demonstrate that the headline configuration owes nothing to one particular split.

Algorithm 1 states the full protocol as pseudocode. Its defining feature is that signals are computed once, outside every grid-search and ablation loop. Separating signal computation from the decision-agent grid search in this way cuts end-to-end runtime by better than an order of magnitude, and it renders the sweep in Section 5.1.9 inexpensive, since altering a cost obliges no agent signal to be recomputed.

5.1.4. Baselines

Baselines fall into three families. *Index buy-and-hold* comes first: the initial cash buys the WIG20 index on the opening date of the test window and the position is held to the end of it, with the FTSE 100 index substituted under the same rule for the replication. Evaluations of trading strategies conventionally adopt buy-and-hold in this role [14].

The second family consists of *single-agent* runs, in which the decision agent is fed by one signal agent alone, giving five baselines, one for each indicator. Five *leave-one-agent-out* configurations accompany them, each withdrawing a single agent from the otherwise complete ensemble. The third family holds the two *machine-learning baselines* that Section 5.1.5 describes.

On each market every baseline is measured over the same test window, charged the market-

Algorithm 1 Anchored walk-forward with precomputed signals

```
1: Input: prices  $P$ ; agent set  $\mathcal{A}$ ; modes  $\mathcal{M}$ ; thresholds  $\mathcal{C}$ ; train/test split  $(T_{\text{tr}}, T_{\text{te}})$ ; costs  $(\gamma, s)$ .
2: Output: selected  $(m^*, c^*)$ ; test-window metrics.
3:  $S \leftarrow \text{PrecomputeSignals}(P, \mathcal{A})$  {all dates, all symbols}
4:  $\text{best} \leftarrow -\infty$ 
5: for all  $m \in \mathcal{M}, c \in \mathcal{C}$  do
6:    $E_{\text{tr}} \leftarrow \text{Backtest}(S, T_{\text{tr}}, m, c, \gamma, s)$ 
7:    $r \leftarrow \text{TotalReturn}(E_{\text{tr}})$ 
8:   if  $r > \text{best}$  then
9:      $\text{best} \leftarrow r; (m^*, c^*) \leftarrow (m, c)$ 
10:  end if
11: end for
12:  $E_{\text{te}} \leftarrow \text{Backtest}(S, T_{\text{te}}, m^*, c^*, \gamma, s)$ 
13: return  $(m^*, c^*), \text{Metrics}(E_{\text{te}})$ 
```

specific transaction costs of Sections 5.1.2 and 5.3.7, and run at the decision-agent configuration that the walk-forward protocol selected. Two exceptions apply: buy-and-hold carries no decision-agent parameters at all, and the machine-learning baselines put a learned classifier where the decision agent would otherwise sit.

5.1.5. Machine-learning baselines

A directional LSTM classifier after the manner of [4] supplies a contemporary learning-based reference point. Per symbol it reads a rolling window of causal daily features drawn from the same price history the agents see, namely multi-horizon returns, realised volatility, and normalised versions of the RSI, ROC, MACD-histogram, Bollinger %b and moving-average-spread indicators, and from these predicts the probability that the return h days ahead is positive. Architecturally it is one LSTM layer followed by dropout and a linear output unit, trained under binary cross-entropy with the Adam optimiser.

Fairness of the comparison requires that every model-selection decision be taken strictly in sample, and it is. The prediction horizon h , the operating threshold and the early-stopping point are all chosen on an internal validation slice cut chronologically out of the training window, and no tuning ever touches the held-out test window. Standardisation statistics for the features are estimated on the training window alone, which leaves the procedure free of look-ahead.

Converting per-symbol probabilities into long or flat positions goes through a symmetric no-trade band about 0.5. Anything above $0.5 + \delta$ opens a long position or keeps one open, anything below $0.5 - \delta$ closes it, and probabilities falling between the two leave the current position as it stands. The arrangement imitates the hold behaviour of the decision agent and heads off the excessive turnover that re-predicting daily would otherwise generate. In-sample selection settles on a horizon of $h = 3$ days and a band of $\delta = 0.03$ on both markets. The long and flat signals that result run through the same portfolio simulator at identical costs and position sizing, so that whatever difference emerges belongs to the decision rule and not to the accounting.

Second among the machine-learning baselines is a gradient-boosted-trees (GBM) directional classifier [28], built on LightGBM [29]. Large-scale empirical comparisons place gradient-boosted trees consistently among the strongest learners available for tabular financial features [37, 38], which makes them a natural complement to the recurrent LSTM and structurally quite unlike it.

The GBM reads, for every symbol and day, the same causal daily features the LSTM reads, extended by lagged copies of each feature out to 20 trading days, which gives the flat feature vector a temporal context comparable to the LSTM lookback window. Training pools the feature rows across symbols, divides them chronologically into training and validation slices, stops early on validation log-loss, and averages three models grown from different seeds. As with the LSTM, the prediction horizon and the no-trade band are fixed strictly in sample, here by the net-of-cost Sharpe ratio of a portfolio simulation on the validation slice, and the long and flat signals produced pass through the same portfolio simulator under identical costs. In-sample selection yields $h = 1$ with $\delta = 0.03$ on WIG20, and $h = 3$ with $\delta = 0.03$ on the FTSE 100.

5.1.6. Risk-adjusted metrics

Metrics are computed from the equity curve and from the list of closed trades, as follows. **Total return** gives the percentage change in portfolio value between the first and the last date of the evaluation window. **CAGR** (compound annual growth rate) annualises that return over the same window. **Max drawdown** takes the largest peak-to-trough decline and reports it as a negative fraction. **Sharpe ratio** [19] divides annualised excess return by standard deviation, annualising with a factor of $\sqrt{252}$ and assuming a risk-free rate of zero.

Sortino ratio [20] substitutes downside deviation into that denominator, so that only below-target returns are penalised:

$$\text{Sortino} = \frac{\sqrt{252} \bar{r}}{\sqrt{\frac{1}{N} \sum_{t=1}^N \min(r_t, 0)^2}}, \quad (5.2)$$

with $\bar{r}$ the mean daily return and the denominator the downside deviation. **Calmar ratio** [21] divides CAGR by the absolute value of the maximum drawdown. **Ulcer Index** [22] takes in the whole drawdown path rather than its extremum:

$$\text{UI} = 100 \cdot \sqrt{\frac{1}{N} \sum_{t=1}^N \text{dd}_t^2}, \quad (5.3)$$

in which dd_t denotes the running drawdown of the equity curve at date t . **Martin ratio** [22] divides the annualised excess return by the Ulcer Index.

Prolonged drawdowns register more strongly on the Ulcer and Martin statistics than on the peak-to-trough maximum, which is why they serve here as tie-breakers between strategies whose peak drawdowns are alike but whose drawdown profiles are not. **Number of trades** counts the closed trades, and **win rate** gives the fraction of them that realised a positive profit.

5.1.7. Market-regime labelling

Comparing the ensemble with buy-and-hold on pooled aggregates alone would conceal a good deal, so every trading day of the test window is assigned to one of three regimes, *bull*, *correction* or *sideways*. Nothing beyond information available from the WIG20 index itself up to date t enters the assignment, which keeps it clear of future prices.

Write high_t for the rolling 252-day maximum of the WIG20 closing price at t , and ret_t for the price relative $\text{close}_t / \text{high}_t$. A *correction* day is one where $\text{ret}_t < 1 - 0.10$, meaning the index sits at least ten percent under its trailing one-year high. A *bull* day is one where $\text{ret}_t \geq 1 - 0.05$, meaning the index lies within five percent of that high, and where the 50-day SMA of the WIG20 index stands above its 200-day SMA. Whatever satisfies neither condition is *sideways*.

Both thresholds are standard in the practitioner literature on market regimes, and neither was tuned on the training data. Applied to the test window, the labelling returns 52.87% bull days, 21.38% correction days and 25.75% sideways days (Table 5.5), so that all three regimes are represented, and the ensemble faces each one.

5.1.8. Volatility-matched comparison

What follows is an analytical normalisation rather than a directly implementable trading result, and its object is to bring the ensemble and the benchmark onto one risk budget so that their returns may be compared on equal terms. The intuition is straightforward. Spending many days out of the market leaves the ensemble running at lower volatility than a benchmark that is always fully invested, so a bare comparison of cumulative returns penalises it for the risk it declined to take. Removing that confound means multiplying the ensemble returns by a single constant chosen to equate the ex-ante volatilities of strategy and benchmark, and comparing the two equity curves thereafter.

The formal construction begins from a standard criticism of the Sharpe ratio, that its scale invariance measures return per unit of risk while saying nothing about how much risk is being run [13]. Two strategies may each post a Sharpe ratio of 1.0 and yet differ in cumulative return if one realises 10% annualised volatility and the other 20%, for no reason beyond the second taking more risk.

The remedy is to scale the equity curve of the ensemble, ex ante, onto the realised training-window volatility of buy-and-hold WIG20. That target is the annualised standard deviation of WIG20 daily returns over the training window, $\sigma^{\text{target}} = \sqrt{252} \text{std}(r_{\text{train}}^{\text{WIG20}}) = 0.22$, set against a strategy volatility over the same window of $\sigma^{\text{strat}} = \sqrt{252} \text{std}(r_{\text{train}}^{\text{ensemble}}) = 0.11$. Their quotient gives the leverage factor $\lambda = \sigma^{\text{target}} / \sigma^{\text{strat}} = 2.06$, held constant across the test window and applied to the out-of-sample ensemble return. Since λ derives entirely from training-window data, the comparison is genuinely ex ante and never consults out-of-sample returns.

Two cautions attach to it. The primary finding remains the unleveraged ensemble, with the volatility-matched figures serving as a counterfactual on risk budget and not as a recommendation to trade with leverage. And the leverage factor required, $\lambda = 2.06$, exceeds what most retail investors on the WSE could realistically obtain or would prudently employ, so these results are best treated as an analytical ceiling on risk-normalised return, well outside what practice would allow. Section 5.2.4 reports the scaled equity curve together with its metrics.

5.1.9. Cost-sensitivity sweep

Beyond the baseline calibration of 39 bps commission with 5 bps slippage, the transaction-cost parameters are swept across a grid

$$\begin{aligned} \text{commission} &\in \{0, 10, 20, 39, 60, 80, 100\} \text{ bps}, \\ \text{slippage} &\in \{0, 5, 10, 20\} \text{ bps}, \end{aligned} \tag{5.4}$$

with the walk-forward backtest re-run at every point on it. What matters is that the agent signals are computed once and handed to each backtest, so that nothing but the portfolio-level accounting varies across the sweep. The experiment exists to establish what portion of the advantage the ensemble holds over buy-and-hold on the risk axis is owed to the particular cost calibration chosen. Section 5.2.6 presents the complete grid.

5.1.10. Bootstrap confidence intervals and tests

Put intuitively, a bootstrap draws the observed daily returns repeatedly, at random and with replacement, recomputes on each redrawn series whichever metric is in question, and reads off from the spread of the results how far that metric might have moved by chance alone. No assumption that returns follow a normal distribution is required anywhere in the procedure.

Every point estimate of total return and Sharpe ratio carries a 95% percentile bootstrap confidence interval, computed on the vector of daily returns of the equity curve over 10,000 resamples drawn with replacement. Comparisons with the local index buy-and-hold benchmark, with the best single-agent baseline and with the machine-learning baselines of Section 5.1.5 use a paired bootstrap instead: each resample draws one set of dates and applies it to both series, the difference in the statistic of interest is taken, and the one-sided p -value gives the proportion of resamples in which that difference is non-positive. A p -value under 0.05 counts as evidence that the strategy dominates the baseline on the test window.

A bootstrap suits this setting because trading-strategy metrics are known to be distributed non-Gaussianly and path-dependently, and prior work has recommended bootstrap methods precisely here [15]. As the literature on trading-strategy evaluation would lead one to expect, these tests are not anticipated to carry the power needed to detect moderate effect sizes over a single test window of just under two years [12]. The p -values are consequently presented beside the point estimates and confidence intervals, and are not leaned on as the sole arbiter of performance.

5.2. Results on the WIG20 panel

This section reports the results obtained on the primary panel. It begins with the parameter selection performed on the training window, then reports the headline and risk-adjusted comparisons against buy-and-hold on the held-out window, decomposes them by market regime, and closes with two experiments that examine the internal mechanism of the ensemble: the empirical correlations between agent signals, and an audit of the confidence heuristic.

5.2.1. Parameter selection on training data

In-sample total return for every pairing of decision-agent mode with minimum confidence is set out in Table 5.1. The *passive* mode seldom reaches its demanding buy and sell thresholds and, at every value of the minimum confidence, executes no trades whatsoever. The *balanced* mode trades in moderation and returns small positive figures at all thresholds. The highest returns overall belong to the *aggressive* mode, whose best training-window pairing, (*aggressive*, $c_{\min} = 0.05$), becomes the configuration carried forward to out-of-sample evaluation.

This accords with the familiar observation that on a trend-rich daily dataset a moderate tolerance for signals tends to maximise gross return, though the transaction-cost model of Section 5.1.2 takes a noticeable bite out of the net figure.

5.2.2. Headline results on held-out data

Headline metrics for the selected configuration and for buy-and-hold WIG20, computed on the held-out test window, are collected in Table 5.2. Every figure is net of the transaction costs specified in Section 5.1.2.

The benchmark wins on gross total return, taking 0.31 across 435 trading days against 0.23 for the ensemble (CI $[-0.10, 0.68]$), with a one-sided bootstrap p -value of 0.636 on the

Table. 5.1. In-sample total return (train window, 2021-2024) by decision-agent mode and minimum confidence. Higher is better. Selected configuration in bold.

$c_{\min}$	passive	balanced	aggressive
0.00	0.00	0.05	0.48
0.05	0.00	0.09	0.55
0.10	0.00	0.09	0.49
0.15	0.00	0.11	0.40
0.20	0.00	0.23	0.28
0.25	0.00	0.16	0.25
0.30	0.00	0.28	0.23

hypothesis that the ensemble exceeds it.

Shift the comparison onto the risk axis and it inverts. The ensemble records a Sharpe ratio of 1.06 (CI $[-0.43, 2.57]$) where buy-and-hold records 0.82, at a p -value of 0.297. Maximum drawdown comes in at -0.06 against -0.17 , better than a twofold reduction, and the Calmar ratio at 2.10 against 0.96, better than twice as large. The two equity curves are plotted in Fig. 5.1, and Fig. 5.2 lays their underwater curves over one another.

Table. 5.2. Out-of-sample metrics on the held-out test window, net of transaction costs. Brackets give 95% bootstrap confidence intervals.

Metric	Multi-agent system	Buy-and-hold WIG20
Total return	0.23 $[-0.10, 0.68]$	0.31
CAGR	0.13	0.17
Max drawdown	-0.06	-0.17
Sharpe ratio	1.06 $[-0.43, 2.57]$	0.82
Calmar ratio	2.10	0.96
Number of trades	34	–
Win rate	0.65	–

5.2.3. Risk-adjusted comparison

Extending the set of risk-adjusted measures, as Table 5.3 does, only strengthens that inversion. Downside deviation falls from 0.15 for buy-and-hold to 0.08 for the ensemble, a drop of more than forty percent, which carries the Sortino ratio up from 1.19 to 1.60.

Because the Ulcer Index registers how deep drawdowns run and how long they persist, rather than only how deep the worst one goes, it separates the two series more sharply still: 2.39 for the ensemble against 6.69 for buy-and-hold, a reduction approaching two-thirds. The Martin ratio moves correspondingly, from 2.49 to 5.29, better than a factor of two. Rolling 60-day Sharpe ratios for both series are plotted in Fig. 5.3, where the ensemble sits above the benchmark across most of the test window and conspicuously so through the correction subperiods.

5.2.4. Volatility-matched comparison

Comparing the two series on gross total return is systematically unfair to the ensemble, since they do not run at comparable volatilities. Measured on the training window, the ensemble annualises

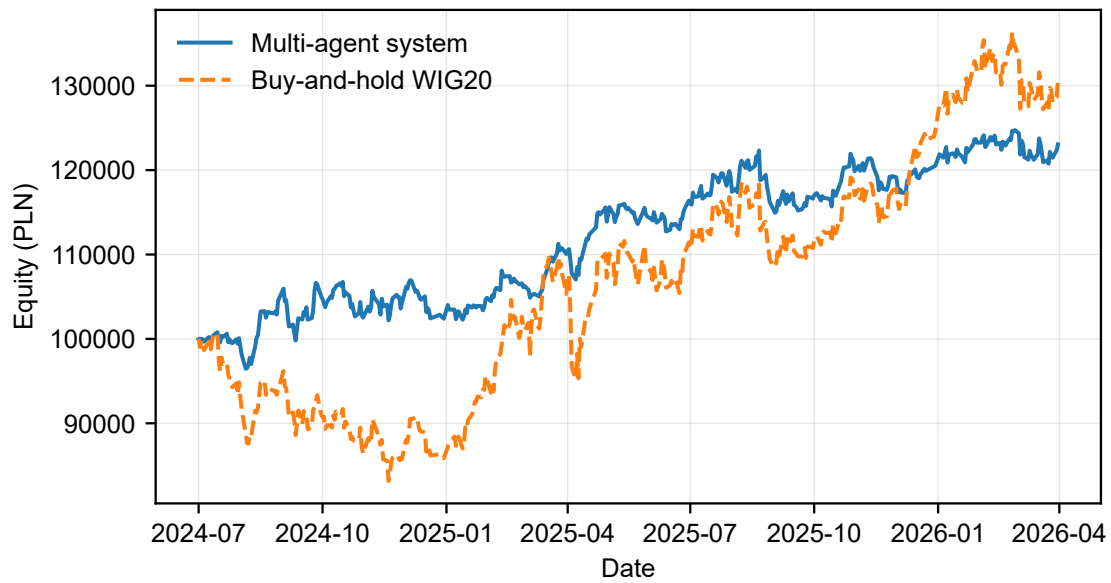


Fig. 5.1. Equity curves on the held-out test window for the multi-agent system (selected configuration) and for buy-and-hold WIG20. Both series start at the same initial cash and are net of transaction costs.

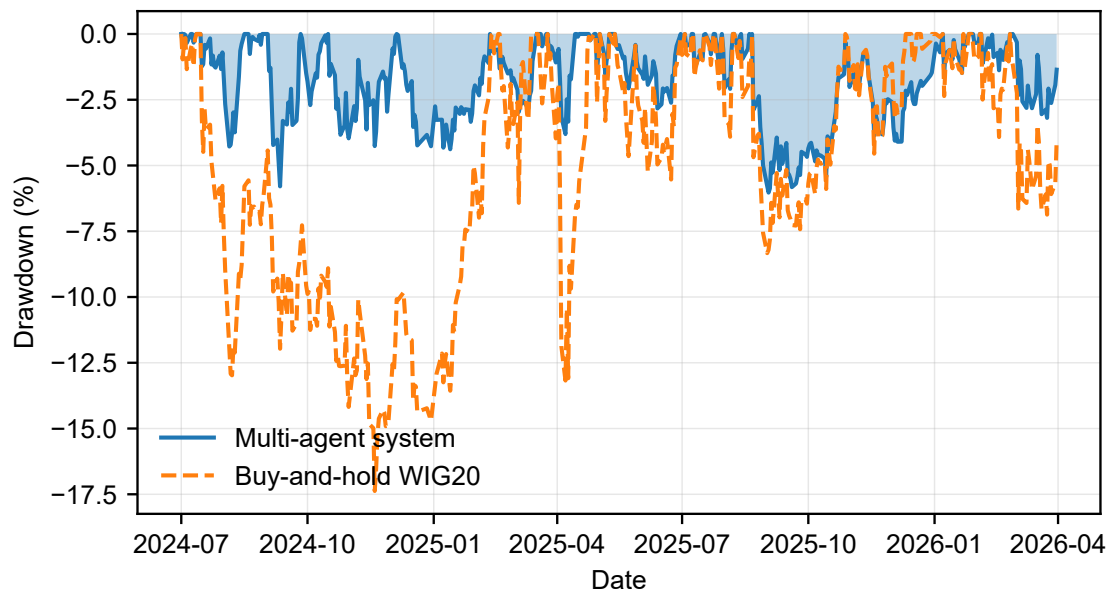


Fig. 5.2. Underwater (drawdown) curves for the ensemble and for buy-and-hold WIG20 on the held-out test window. The drawdowns of the ensemble are both shallower and shorter, which is the central risk-adjusted finding of the thesis.

Table. 5.3. Risk-adjusted metrics on the held-out test window, net of transaction costs.

Metric	Multi-agent system	Buy-and-hold WIG20
Sharpe ratio	1.06	0.82
Sortino ratio	1.60	1.19
Downside deviation	0.08	0.15
Calmar ratio	2.10	0.96
Max drawdown	−0.06	−0.17
Ulcer Index	2.39	6.69
Martin ratio	5.29	2.49

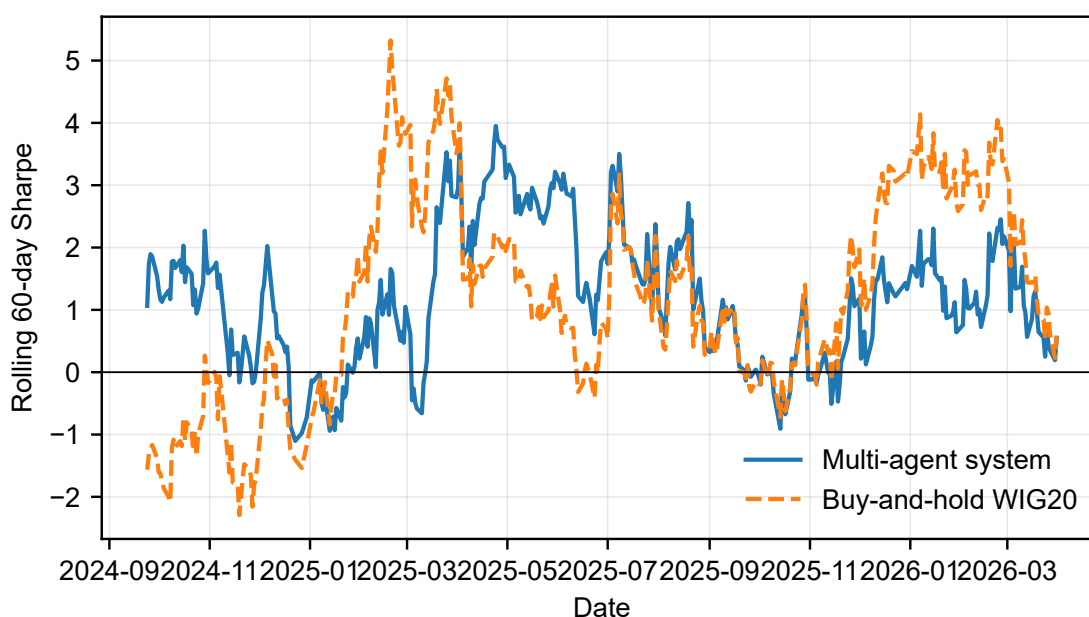


Fig. 5.3. Rolling 60-day Sharpe ratio of the ensemble and of buy-and-hold WIG20 on the held-out test window. The rolling Sharpe of the ensemble exceeds buy-and-hold during correction subperiods and is close to it during the bull leg.

at 0.11 and buy-and-hold WIG20 at 0.22. Equalising the two risk budgets means multiplying its out-of-sample daily returns by the ex-ante leverage factor $\lambda = 2.06$ that Section 5.1.8 constructs.

Table 5.4 collects what results. Matched on volatility, the ensemble returns 0.49 where buy-and-hold returns 0.31, an advantage of +18.55 percentage points. Sharpe is unmoved at 1.06, necessarily so, since ignoring cash reserves makes it scale-invariant in the leverage factor. Sortino comes to 1.63 against 1.19, and Calmar to 2.13 against 0.96. Maximum drawdown under matching is −0.12, which remains materially below the −0.17 of buy-and-hold notwithstanding the far greater leverage, because the correction-regime drawdown of the ensemble is structurally shallower to begin with (Section 5.2.5).

These figures are analytical instruments rather than trading instructions. Leverage of magnitude $\lambda = 2.06$ lies beyond what a typical retail investor on the WSE can practically obtain, so what Table 5.4 offers is a counterfactual on risk budget, not an outcome anyone could realise. The purpose is corrective: measuring a partially invested strategy at its natural volatility against a fully invested benchmark running at roughly twice that volatility understates how much return per unit of risk the strategy actually delivers.

Table. 5.4. Volatility-matched out-of-sample metrics. The ensemble is scaled, ex ante, to the buy-and-hold volatility measured on the training window (leverage factor $\lambda = 2.06$).

Metric	Ensemble (unleveraged)	Ensemble (vol-matched)	Buy-and-hold WIG20
Total return	0.23	0.49	0.31
CAGR	0.13	0.26	0.17
Sharpe ratio	1.06	1.06	0.82
Sortino ratio	1.60	1.63	1.19
Max drawdown	−0.06	−0.12	−0.17
Calmar ratio	2.10	2.13	0.96
Ulcer Index	2.39	4.96	6.69

5.2.5. Market-regime decomposition

Three markedly different market regimes are blended together in the aggregates of Table 5.2. Under the labelling of Section 5.1.7 the test window holds 230 bull days (52.87% of it), 93 correction days (21.38%) and 112 sideways days (25.75%). Table 5.5 separates the total return and the Sharpe ratio of the ensemble and of buy-and-hold WIG20 across those three.

Through the corrections the ensemble gives up −0.07 where buy-and-hold gives up −0.20, a defensive margin of +12.57 percentage points. Through the sideways stretches it takes 0.05 against −0.00 for the benchmark, an excess of +5.14 percentage points. Through the bull phases the ordering reverses, 0.26 against 0.63, a shortfall of −36.84 percentage points.

That accounting reconciles the aggregate numbers of Table 5.2. Full exposure through the bull leg lets buy-and-hold collect most of its upside, whereas the ensemble deliberately reduces exposure across the correction and sideways regimes and surrenders part of the bull-regime upside as the price. Where a bull regime dominates a test window, aggregate total return will favour the fully invested benchmark mechanically; give the window a larger correction share and the aggregate comparison would swing to the ensemble.

Table. 5.5. Decomposition of the held-out test window into bull, correction, and sideways subperiods. Shares refer to the fraction of trading days in the test window.

Regime	Days	Share	Ens. TR	Ens. Sharpe	B&H TR	B&H Sharpe
Bull	230	52.9%	0.26	2.47	0.63	2.76
Correction	93	21.4%	−0.07	−1.34	−0.20	−2.41
Sideways	112	25.7%	0.05	0.93	−0.00	0.11
All	435	100.0%	0.23	1.06	0.31	0.82

5.2.6. Cost-sensitivity analysis

Out-of-sample Sharpe ratios across the cost grid of Section 5.1.9 are tabulated in Table 5.6. Charging nothing yields 1.15; the baseline calibration of 39 bps commission with 5 bps slippage yields 1.06; the harshest corner of the grid, 100 bps commission with 20 bps slippage, brings it down to 0.92. Nowhere on the grid does the ratio fall to the benchmark level, its minimum of 0.92 still exceeding the buy-and-hold Sharpe of 0.82.

Total return, computed alongside Sharpe but omitted from the table for brevity, tells the same story qualitatively. At every point on the grid the ensemble trails buy-and-hold on gross

total return, and at every point it posts a higher Sharpe ratio, a higher Sortino ratio and a smaller drawdown. The risk-adjusted edge is thus not manufactured by one particular cost calibration, but is a stable property of the strategy across the plausible range of retail brokerage fees.

Table. 5.6. Cost sensitivity of the out-of-sample Sharpe ratio. Rows are commission in basis points; columns are one-way slippage in basis points.

Commission (bps)	0 bp	5 bp	10 bp	20 bp
0	1.15	1.14	1.13	1.11
10	1.13	1.12	1.11	1.09
20	1.11	1.10	1.09	1.07
39	1.07	1.06	1.05	1.03
60	1.03	1.02	1.02	0.99
80	0.99	0.99	0.98	0.96
100	0.96	0.95	0.94	0.92

5.2.7. Empirical signal correlations

Everything the variance-reduction argument of Section 3.1 claims depends on the five agents producing signals whose errors are imperfectly correlated. Table 5.7 puts that dependence to the test, giving the pairwise Pearson correlations [26] of the signed confidence values over all pooled (symbol, date) observations in the training window.

Over 10 distinct pairs the mean absolute off-diagonal correlation is 0.25, bounded by -0.71 at one end (RSI against ROC) and 0.46 at the other (RSI against Bollinger). The momentum family (MACD, ROC) and the mean-reversion family (RSI) correlate negatively, which follows from their reading the same raw price input in opposite directions, and the trend agent (MA trend) comes out close to uncorrelated with everything else, which follows from its much longer horizon.

The same matrix is rendered as a heatmap in Fig. 5.4. What these measurements supply is the non-identical-errors assumption of (3.1) in concrete form, and Section 5.4 draws on them again in explaining why most of the variance reduction comes from the interplay of the momentum and mean-reversion families rather than from the trend agent acting on its own.

Table. 5.7. Pearson correlations of the pooled (symbol, date) signed confidence values of the five signal agents, computed on the training window. Upper triangle shown.

	MACD	RSI	ROC	Bollinger	MA trend
MACD	1.00	-0.01	+0.04	-0.34	-0.04
RSI		1.00	-0.71	+0.46	+0.12
ROC			1.00	-0.55	-0.12
Bollinger				1.00	+0.09
MA trend					1.00

5.2.8. Validation of the confidence measure

The justification offered for the score (3.5) in Section 3.2 treats per-signal confidence as a heuristic stand-in for inverse forecast variance. Were that reading literally true, it would carry an observable

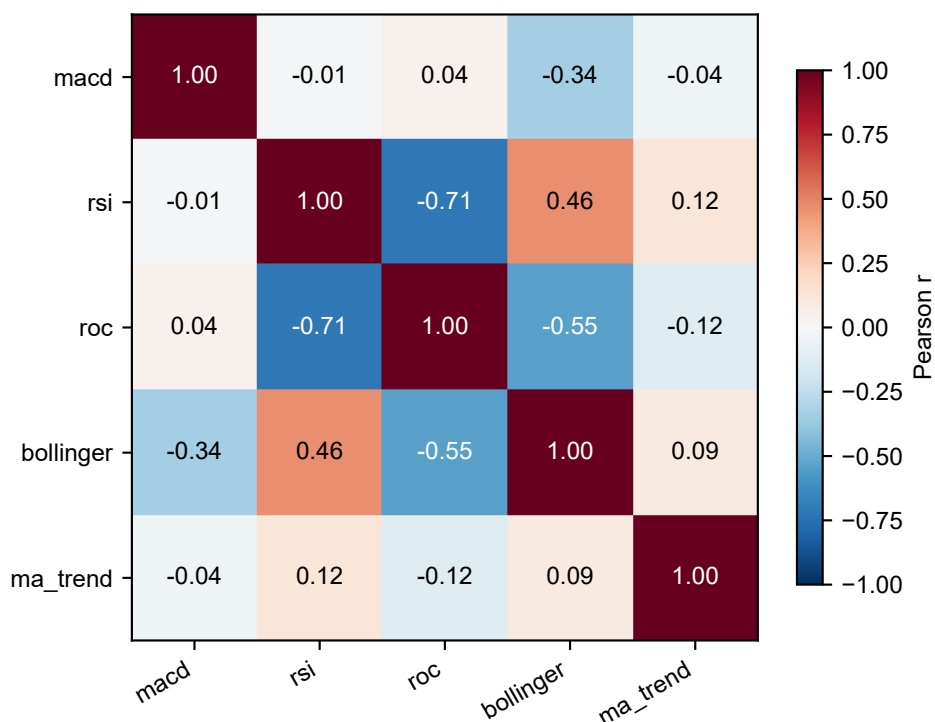


Fig. 5.4. Heatmap of the pairwise Pearson correlations of the agent signed-confidence signals on the training window. Blue cells mark negative correlations, red cells mark positive ones. The largest absolute correlations appear in the RSI-ROC and RSI-Bollinger pairs, which is consistent with their shared reliance on overbought and oversold rules.

consequence: signals issued at higher confidence ought to get the direction right more often than signals issued at lower confidence.

Table 5.8 puts that consequence to the test on the training window, so that no test-window information is used. Every active signal, meaning every signal that is not a hold, is pooled per agent across symbols and dates; its directional hit rate is computed against the following day's move; and confidence is then set against correctness twice over, by a Spearman rank correlation [27] and by weighing the hit rate of the top confidence quintile against that of the bottom, after the manner of the reliability analysis in [54]. Rank correlation is the right instrument here because correctness takes only two values and the confidence distributions pile up heavily towards zero, which leaves no warrant for assuming linearity. The exercise complements rather than repeats the Pearson correlations of Section 5.2.7 [26], which concern the association *between* agents, not that between the confidence of an agent and its own accuracy.

There is no ambiguity in the answer. The confidence values hold no usable information whatever about accuracy at the level of individual signals. Pooling 29,063 training-window signals gives a hit rate of 0.498; the per-agent rank correlations occupy $[-0.035, 0.016]$ and not one of them exceeds $|\rho| = 0.035$; and the top confidence quintile shows no systematic accuracy advantage over the bottom one.

One entry in Table 5.8 is nominally significant (RSI, $p = 0.031$), and its sign is negative, so that greater confidence goes together with marginally *worse* calls. It accounts for well under one percent of the variance in correctness and does not withstand a multiple-testing correction over the ten agent-horizon combinations examined, which makes it evidence against the inverse-variance reading rather than for it. Lengthening the horizon to five days changes nothing (pooled hit rate

Table. 5.8. Confidence validation on the WIG20 training window at the one-day horizon. ρ is the Spearman rank correlation between confidence and directional correctness, with its p -value; Hit (Q1) and Hit (Q5) are hit rates in the bottom and top confidence quintiles.

Agent	n	Hit rate	ρ	p	Hit (Q1)	Hit (Q5)
MACD	1,405	0.472	+0.002	0.955	0.470	0.484
RSI	3,847	0.497	−0.035	0.031	0.525	0.497
ROC	11,155	0.498	+0.016	0.082	0.494	0.520
Bollinger	1,674	0.483	−0.004	0.867	0.478	0.481
MA trend	10,982	0.505	−0.004	0.690	0.503	0.500
Pooled	29,063	0.498	+0.003	0.578	0.493	0.495

0.510), and the FTSE 100 training window returns the same verdict (largest per-agent $|\rho| = 0.027$, pooled hit rate 0.502).

Two consequences follow. The first is that reading the confidence values as inverse variances is a conceptual analogy and not an empirically calibrated property of the system. Taken one signal at a time and at short horizons, the agents behave close to uninformed forecasters, and the magnitude of the confidence attached does nothing to separate the good calls from the bad. The low tuned threshold ($c_{\min} = 0.05$) fits that picture, functioning in practice as a mild noise filter rather than as a precision weight.

The second is that precision-weighted averaging cannot be credited with the risk-adjusted performance recorded in the preceding subsections. Two other things are doing the work: the pooling of signals that correlate weakly or negatively across the methodological families (Table 5.7), which depresses the variance of the aggregate position whether or not the individual confidence magnitudes are informative, and the regime exposure profile that thresholding the score induces (Table 5.5). What this finding implies for the theoretical framing of the thesis is taken up in Section 5.4.

5.3. Robustness checks and external validation

The results of Section 5.2 rest on one configuration, one weighting scheme, one chronological split, and one market. This section removes each of those supports in turn. It reports the single-agent and leave-one-agent-out baselines, the sensitivity of the results to the family weights, the comparison against the two machine-learning baselines, the significance analysis, the three-fold walk-forward, the sensitivity to the selection objective, and the full replication of the pipeline on the FTSE 100.

5.3.1. Single-agent baselines and leave-one-out ablation

Total return and Sharpe ratio for each of the five single-agent runs, alongside those of the full ensemble, appear in Table 5.9. `only_rsi` heads the single-agent field at 0.16 total return. Testing the ensemble against that best single agent on total return returns a one-sided bootstrap p -value of 0.268. H2 rests principally on these figures.

Table 5.10 records what happens when each signal agent is withdrawn in turn. Every removal registers somewhere in the headline metrics, so no agent is wholly idle, yet in each case the reduced ensemble still leads buy-and-hold WIG20 on point estimates of Sharpe ratio and of maximum drawdown. H3 rests principally on this.

Table. 5.9. Single-agent baselines on the test window, with the selected mode and minimum confidence. All numbers net of costs.

Agent used	Total return	Sharpe ratio
MACD only	0.15	0.86
RSI only	0.16	1.03
ROC only	0.00	0.13
Bollinger only	0.00	0.00
MA trend only	−0.00	0.03
Full ensemble	0.23	1.06

Table. 5.10. Leave-one-agent-out ablation on the test window. The variant column indicates which agent was removed.

Variant	Total return	Sharpe ratio
Full ensemble	0.23	1.06
no MACD	0.28	1.28
no RSI	0.23	0.92
no ROC	0.26	1.17
no Bollinger	0.23	1.06
no MA trend	0.19	1.16

5.3.2. Weight sensitivity and alternative weighting schemes

Since w_{κ} constitutes the only fixed prior anywhere in the decision layer, how far the headline results lean on it deserves measurement. Throughout this subsection the walk-forward-selected configuration (aggressive, $c_{\min} = 0.05$) is held constant and the weights alone are varied, so that whatever moves can be attributed to the weighting scheme and to nothing else.

Three families of alternatives are put on the held-out window. The first perturbs one family weight at a time by $\pm 25\%$ and $\pm 50\%$, yielding 16 variants. The second abandons the ordering rationale of Section 3.6 outright, in two ways: weights made equal across the families, and weights reversed so that volatility ranks highest and trend lowest. The third estimates the weights from the training window alone, again in two ways. *Optimised* weights come from an exhaustive grid over all four families, judged by the same in-sample total-return objective the headline protocol uses. *Inverse-variance* weights are set proportional to the reciprocal of the directional error variance of each family, as measured on training-window signal outcomes, which is the classical prescription of [23] made operational. The named schemes are reported for WIG20 in Table 5.11.

Three conclusions emerge. The first is that the weights are not what carries the risk-adjusted profile. Out-of-sample Sharpe ratios across the 16 perturbations occupy $[0.70, 1.38]$ and total returns occupy $[0.14, 0.30]$, while just 2 of the 20 fixed and theory-implied schemes finish beneath the buy-and-hold Sharpe ratio of 0.82. Perturbing the volatility weight changes nothing whatsoever in any of its four variants, for the simple reason that the selected configuration leaves the Bollinger agent essentially inactive.

With hindsight the baseline is not the best available configuration: cutting the momentum-family weight moderately would have served this particular window better, which agrees with the ablation result that dropping MACD helps. The two schemes falling under the benchmark

Table. 5.11. Alternative weighting schemes on the WIG20 test window, with (mode, $c_{\min}$) fixed at the walk-forward selection. Weights are listed as (trend, momentum, mean-reversion, volatility).

Scheme	Weights	TR	Sharpe	MDD
Baseline	(1.3, 1.0, 0.8, 0.5)	0.23	1.06	−0.06
Equal	(1.0, 1.0, 1.0, 1.0)	0.25	1.16	−0.06
Inverted	(0.5, 0.8, 1.0, 1.3)	0.23	1.23	−0.05
Optimised	(1.30, 0.80, 0.80, 0.50)	0.25	1.18	−0.05
Inv.-variance	(1.02, 1.00, 1.00, 0.98)	0.25	1.16	−0.06
Buy-and-hold	–	0.31	0.82	−0.17

Sharpe ratio do forfeit that portion of the risk-adjusted advantage, but even they hold the maximum drawdown well inside the buy-and-hold −0.17. Drawdown reduction, the central risk-adjusted finding of this thesis, thus outlives every weighting scheme tried, including those where the Sharpe advantage does not.

The second conclusion is that fitting the weights in sample gains little. Grid optimisation returns almost the fixed prior back again, moving only the momentum weight, from 1.0 to 0.80, and its out-of-sample total return of 0.25 sits roughly two percentage points from the baseline. The FTSE 100 goes further and has the optimised weights lose to the fixed prior out of sample (0.13 against 0.14 on total return). That is the forecast combination puzzle in miniature, with the estimation error in the fitted weights consuming the gains they appeared to offer [50, 51].

The third conclusion is that the inverse-variance prescription has nothing to say in this setting. Directional hit rates barely differ between the families (Section 5.2.8), so the estimation delivers weights that are nearly equal (1.02, 1.00, 1.00, 0.98 for trend, momentum, mean reversion and volatility), and performance accordingly matches the equal-weight scheme almost exactly.

Repeating the whole experiment on the FTSE 100 reproduces the qualitative pattern, with perturbation Sharpe ratios spanning [0.69, 1.54] and 3 of 20 fixed schemes under the buy-and-hold Sharpe ratio. Set beside the leave-one-agent-out ablation, these results say that what the ensemble contributes does not depend on the particular weight values chosen, and that the coarse fixed prior holds up against perturbed, reordered, optimised and theory-implied alternatives alike.

5.3.3. Comparison with machine-learning baselines

Table 5.12 sets the ensemble against the two machine-learning baselines of Section 5.1.5 over the held-out WIG20 window. The LSTM finishes at 0.08 total return and a Sharpe ratio of 0.44, where the ensemble reaches 0.23 and 1.06 and buy-and-hold 0.31 and 0.82. The ensemble is ahead on total return, on Sharpe ratio and on maximum drawdown (−0.06 against −0.11), and it gets there on a far smaller number of trades (34 against 237). One-sided paired bootstrap p -values for the ensemble exceeding the LSTM come to 0.068 on total return and 0.063 on Sharpe ratio, so, just as in the buy-and-hold comparison, a sizable point-estimate advantage still falls short of the conventional $p < 0.05$ threshold on a single test window.

The GBM baseline behaves much the same way, returning 0.04 at a Sharpe ratio of 0.47 across 34 closed trades. Statistically its equity path cannot be separated from that of the LSTM (paired one-sided $p = 0.621$ on total return), and the ensemble again leads on total return, on Sharpe ratio and on Calmar ratio (one-sided $p = 0.107$ on total return and $p = 0.210$ on Sharpe ratio).

Maximum drawdown is the one measure on which the GBM comes out in front (−0.05

against -0.06), and what lies behind that number is a nearly flat equity curve rather than superior risk control. Very little risk is taken and correspondingly little is earned, which the Calmar ratio duly registers (0.45 against 2.10). That two learners as structurally unlike as a recurrent network and a boosted-tree ensemble, each tuned wholly in sample beneath one protocol, should converge on out-of-sample results this modest reinforces the earlier reading. Flexible learners find little to exploit in the WIG20 test window, while the transparent forecast-combination ensemble stays competitive without being trained at all, which accords with the wider finding that simple, well-regularised combinations hold their own against more flexible learners on noisy financial series [24, 25].

Table. 5.12. Comparison of the ensemble, the LSTM and GBM directional baselines, and buy-and-hold on the held-out WIG20 test window, net of transaction costs.

Metric	Multi-agent	LSTM	GBM	Buy-and-hold
Total return	0.23	0.08	0.04	0.31
Sharpe ratio	1.06	0.44	0.47	0.82
Max drawdown	-0.06	-0.11	-0.05	-0.17
Calmar ratio	2.10	0.42	0.45	0.96
Number of trades	34	237	34	–

5.3.4. Statistical significance

Two kinds of evidence are attached to the point estimates by the bootstrap. The first is interval evidence. Percentile confidence intervals at the 95% level express the sampling uncertainty that a test window of 435 trading days necessarily carries, and for the ensemble they run $[-0.10, 0.68]$ on total return and $[-0.43, 2.57]$ on Sharpe ratio. Both buy-and-hold point estimates, 0.31 on total return and 0.82 on Sharpe ratio, fall strictly inside those intervals, which is the statistical way of saying that the hypothesis of a common underlying mean return and Sharpe ratio cannot be rejected on this window.

The second is test evidence. Paired against buy-and-hold, the one-sided p -values come to 0.636 on total return and 0.297 on Sharpe ratio, and neither clears the conventional $p < 0.05$ threshold.

Two readings are available. On the risk axis, meaning drawdown, Sortino, Ulcer and Martin, the test-window point estimates favour the ensemble consistently, by margins that are economically large and hold up reasonably well across the ablation and cost grids. Yet rejecting a null of equal mean performance on a single window of under two years is difficult, because trading-strategy return distributions are heavy-tailed and path-dependent [15, 12]. Standard practice in this literature is therefore followed: point estimates, confidence intervals and p -values are presented together, and no p -value is treated as the sole arbiter. That the buy-and-hold estimates sit inside the bootstrap intervals of the ensemble supports treating the two series as statistically indistinguishable on gross total return, while the large and repeated gap on drawdown-based and downside-based metrics supports the risk-aware reading of what the ensemble contributes (Section 5.4).

Two points bear emphasis if these numbers are to be read correctly. One concerns how wide the intervals are. Total return at $[-0.10, 0.68]$ and Sharpe at $[-0.43, 2.57]$ come out broad because 435 trading days furnish relatively few independent realisations of the quantities being estimated, and because serial dependence and heavy tails together inflate the variance of the resampled statistics. At this sample length the paired bootstrap has limited power, and even a

genuine moderate edge would frequently fail to register at $p < 0.05$. The non-significant p -values are consequently a symptom of a short out-of-sample window, and not positive evidence that the two strategies are equivalent.

The other concerns the gap between practical and statistical significance. What the ensemble achieves on the drawdown-based and downside-based metrics is practically significant, in that the margins are economically large and stable across the ablation, cost and regime analyses, and at the same time not statistically significant, in that no null hypothesis is rejected on this one window. The contribution of the ensemble is framed throughout the thesis accordingly, as consistent and practically meaningful risk reduction, and no claim of statistical dominance over buy-and-hold is advanced anywhere.

5.3.5. Walk-forward robustness

Results of the anchored three-fold walk-forward appear in Table 5.13, giving for every fold both the configuration chosen on its training window and the total return achieved on its test slice. That choice holds steady from fold to fold, which says that the headline configuration is not an artefact of one particular chronological split.

Table. 5.13. Anchored walk-forward with $K = 3$ folds. Selected mode and minimum confidence per fold, plus the test-window total return of each fold.

Fold	Selected mode	$c_{\min}$	Test total return
1	aggressive	0.05	0.05
2	aggressive	0.05	0.09
3	aggressive	0.10	0.08

5.3.6. Sensitivity to the selection objective

Choosing (m^*, c^*) by in-sample total return, as the headline protocol does (Section 5.1.3), raises an obvious question: would a risk-adjusted objective have chosen differently? The test is to re-rank the very same training-window grid under two risk-adjusted objectives, the Sharpe ratio and the Calmar ratio, and then to evaluate each winner untouched on the held-out window.

It would not. On WIG20 all three objectives converge on (aggressive, $c_{\min} = 0.05$), and the FTSE 100 replication behaves the same way, returning (aggressive, $c_{\min} = 0.15$) whichever objective is applied. Substituting a risk-adjusted objective for the total-return one therefore leaves every out-of-sample number reported in this thesis unchanged on both markets.

Table 5.1 shows why. No *passive* configuration trades at all, and the *balanced* ones trade too seldom to compete on cumulative return or on ratio statistics, which leaves the argmax inside the *aggressive* column under every objective. Such invariance should not be assumed in general, as it may well break on grids with more degrees of freedom or on longer menus of modes. On the present design, though, it confirms that opting for a simple, low-variance selection objective (Section 5.1.3) is not load-bearing, which is what the estimation-noise case against ratio objectives on short training samples would predict [53].

5.3.7. External validation on the FTSE 100

Is the risk-aware profile a property of the WSE or of the method? Settling that requires running the whole pipeline again somewhere else, which is what this section does on the FTSE 100 panel of Section 5.1.1. Only two things change. The price data become British, and the cost model is recalibrated to UK conditions: 5 bps of proportional commission, 5 bps of one-way slippage, and a 50 bps Stamp Duty Reserve Tax levied on purchases alone. That last item makes the FTSE cost environment asymmetric and, for buying, appreciably harsher than the Polish calibration. Nothing else moves at all: the decision-agent grid, the indicator hyperparameters and the whole evaluation procedure stay exactly as they were.

Walk-forward selection lands on the same decision-agent family it chose for WIG20, aggressive mode, and the resulting configuration is measured over 444 held-out trading days. Table 5.14 carries the headline comparison, and its shape will be familiar. Gross total return again favours the benchmark (0.14 against 0.25), while every risk measure again favours the ensemble, with a Sharpe ratio of 1.37 against 1.13, a maximum drawdown of -0.07 against -0.13 for a reduction of roughly one half, and a Calmar ratio of 1.18 against 1.00.

The extended measures repeat the pattern: Sortino at 1.99 against 1.52, Ulcer Index at 1.59 against 2.66, and downside deviation at 0.04 against 0.09. Exactly one measure reverses, the Martin ratio (4.91 against 5.05), and it does so narrowly. The explanation lies in the composition of the window: a test period this overwhelmingly bullish leaves buy-and-hold with an integrated drawdown so shallow that a ratio built on it is hard to beat.

Table. 5.14. External validation on the FTSE 100 held-out test window, net of transaction costs (including the 50 bps buy-side Stamp Duty Reserve Tax). Brackets give 95% bootstrap confidence intervals.

Metric	Multi-agent	LSTM	GBM	Buy-and-hold
Total return	0.14 [−0.01, 0.32]	0.27	0.35	0.25
CAGR	0.08	–	–	0.13
Sharpe ratio	1.37 [−0.11, 2.94]	1.49	1.63	1.13
Max drawdown	−0.07	−0.08	−0.10	−0.13
Calmar ratio	1.18	–	1.88	1.00
Sortino ratio	1.99	–	–	1.52
Ulcer Index	1.59	–	–	2.66
Number of trades	55	37	0	–

Decomposing by regime accounts for the aggregates. Bull days make up 90.32% of the FTSE window (401 days out of 444), leaving only 5 correction days and 38 sideways days, so the correction-regime defence responsible for much of the WIG20 result has almost nowhere to act. The sideways carry survives regardless: across those subperiods the ensemble takes 0.02 against -0.03 for buy-and-hold, reproducing the WIG20 pattern. As before, the aggregate return shortfall sits in the bull regime (0.14 against 0.30), where being partially invested mechanically loses to being fully invested. Putting the two series on a common ex-ante volatility, by the construction of Section 5.2.4, lifts the ensemble to 0.26 at a leverage of $\lambda = 1.82$, which is +1.87 percentage points clear of buy-and-hold and still at the shallower drawdown (-0.12).

On statistical power the paired bootstrap repeats the WIG20 story. Buy-and-hold point estimates fall within the ensemble confidence intervals, and the one-sided p -values for the ensemble exceeding the benchmark come to 0.789 on total return and 0.331 on Sharpe ratio, neither of

which reaches $p < 0.05$.

On this window the LSTM baseline outscores the ensemble on point estimates, taking 0.27 against 0.14 on gross total return and 1.49 against 1.37 on Sharpe ratio. Neither margin attains statistical significance, 0.879 being the one-sided p -value on the ensemble exceeding the LSTM in total return. Two observations place the apparent edge in context. It reflects bull-market exposure rather than better risk control, because the LSTM stays invested almost continuously across a window that is 90.32% bull and therefore harvests more of the upside while accepting a deeper drawdown (-0.08 against -0.07). And the LSTM cannot be told apart from passive buy-and-hold on this window either (one-sided $p = 0.411$), which is what a near-fully-invested strategy tracking its index looks like statistically. The transparent ensemble, by contrast, gives up part of that exposure by design in return for the shallower drawdown recorded above.

The GBM baseline carries the same pattern to its limit. It records the best point estimates anywhere on the window, 0.35 total return and a Sharpe ratio of 1.63, and arrives there by closing no position whatsoever across the 444 held-out days, which amounts to a fully invested stance throughout a window that is 90.32% bull. On total return its margin does reach marginal significance at five percent, 0.951 being the one-sided p -value on the ensemble exceeding it, though the matching Sharpe-ratio comparison does not (one-sided $p = 0.682$), and measured against buy-and-hold the advantage stays statistically non-significant (one-sided $p = 0.144$ on total return). All of which is the signature of a near-fully-invested strategy riding a bull market rather than of superior risk control.

Its maximum drawdown, -0.10 , runs deeper than the -0.07 of the ensemble, and so does that of the LSTM, which leaves the qualitative trade-off where it stood. Staying exposed earns the learning-based baselines more of the bull market, and the ensemble spends that upside deliberately, buying drawdown control with it.

The auxiliary analyses transfer to this market as well. Weight sensitivity produces the same qualitative ranking (Section 5.3.2), the walk-forward selection remains invariant to the objective (Section 5.3.6), and the confidence heuristic proves just as uninformative at signal level (Section 5.2.8). The FTSE evidence therefore establishes that the central finding of this thesis, a consistent advantage in drawdown and downside risk over passive buy-and-hold bought with a portion of bull-market upside, carries to a second market whose cost structure differs markedly, and is no artefact of the WIG20 sample.

5.4. Discussion

This section interprets the results reported above. It first traces the path from forecast-combination theory to the empirical evidence, then assesses the three hypotheses in turn, examines how far the findings generalise across markets, explains why a comparison on gross total return understates what the ensemble contributes, and closes with the practical, deployment, and design implications.

5.4.1. From forecast-combination theory to empirical evidence

The prediction carried by (3.1) is that a pool of forecasters whose errors are less than perfectly correlated incurs generally less error than the best of its members. Whether that prediction has any purchase on the present system is an empirical question about the agent pool, and Section 5.2.7 answered it. Off-diagonal correlations among the signed confidence values run from -0.71 to 0.46 and average 0.25 in absolute terms (Table 5.7, Fig. 5.4). Negative pairs count for most in the identity, because a negative ρ_{kj} subtracts from the double sum instead of merely adding little to it,

and two such pairs stand out: RSI against ROC at -0.71 , and Bollinger against ROC at -0.55 .

Of these the RSI-ROC pair is the most instructive, because its negative sign is not an accident of the sample but a consequence of how the two rules are built. Oversold conditions read as an opportunity to a mean-reversion rule and as a warning to a momentum rule, so RSI and ROC are constructed to take opposite sides of the same evidence. Disagreement of that kind is what the score in (3.5) exists to adjudicate, and it does so by weighing the competing confidences under their family priors. Which family prevails shifts with the market: trend carries the score through bull phases, while mean reversion and volatility recover relative influence when the index moves sideways or falls. Table 5.5 records the consequence of that shifting balance, since the ensemble surrenders less than buy-and-hold in corrections and gains more in sideways stretches, which is the profile a variance-minimising pool of heterogeneous forecasters would be expected to produce.

A worked example makes the arithmetic concrete. Take a WIG20 constituent that has run up sharply across the previous ten sessions. ROC, reading nothing but the speed of that move, returns a buy at high confidence. RSI, compressing the same ten sessions into a 14-day relative strength that has now passed 70, returns a sell at moderate confidence. MA trend also returns a buy, since the fast SMA sits above the slow one, but at modest confidence, because the gap between the two averages is small measured against the latest close.

With the family weights fixed as in Section 3.6, those three readings enter (3.5) as $1.0 \cdot c_{\text{ROC}}$ and $1.3 \cdot c_{\text{MA}}$ on the positive side against $-0.8 \cdot c_{\text{RSI}}$ on the negative. The resulting action is settled by the confidences rather than by a vote: let the overbought reading be emphatic enough and the total falls short of the buy threshold, even with two of the three agents asking to go long. That is (3.1) restated in the vocabulary of the decision layer. A disagreement between a momentum rule and a mean-reversion rule keeps the ensemble out of an exposure that either family, left to itself, would have held, and the accumulated effect of such abstentions is the drawdown profile of Fig. 5.2.

Section 5.2.8 then fixes the limits of the analogy. Since the confidence magnitudes prove to hold no measurable information about whether a signal is right, the reading of (3.5) as an inverse-variance weighting survives only as the motivation that shaped the architecture, and not as a property the built system possesses. Forecast-combination theory offers two claims here, and the evidence separates them. The strong claim, that the more reliable forecasters can be identified and weighted up, finds no support. The weak claim, that pooling imperfectly correlated errors shrinks the variance of the combination however the pool is weighted, is what the data bear out.

Section 5.3.2 corroborates that separation from the opposite direction. Equal weights, estimated inverse-variance weights and the fixed prior all land in much the same place out of sample, which is what the combination literature leads one to expect when the pooled members are of comparable individual accuracy [24, 50]. The operative content of the theory in this application is therefore error diversity alone, and every claim made in this thesis is scoped accordingly.

5.4.2. Evidence for the three hypotheses

H1 asserted that the system would match or exceed buy-and-hold on risk-adjusted grounds even where it lost on gross return, and four sets of results bear on it: Tables 5.2, 5.3, 5.4 and 5.5. Their verdict is affirmative on point estimates. The bull-regime shortfall of -36.84 percentage points is by itself more than sufficient to account for the aggregate deficit on gross total return, so the return comparison is settled inside that one regime. On every risk-adjusted measure computed, meanwhile, the ordering runs the other way: Sharpe (1.06 against 0.82), Sortino (1.60 against

1.19), Calmar (2.10 against 0.96), Ulcer Index (2.39 against 6.69), Martin (5.29 against 2.49) and maximum drawdown (−0.06 against −0.17).

Equalising the two risk budgets closes the return gap as well. On the volatility-matched basis the ensemble finishes +18.55 percentage points ahead of buy-and-hold while still drawing down materially less. Every one of these margins is a point estimate, and Section 5.3.4 shows that none attains conventional significance on a single sub-two-year window, so they are read here as consistent and economically meaningful rather than as demonstrated dominance. Section 5.3.7 recovers the same qualitative pattern across 444 held-out trading days under UK-specific costs, which speaks to external validity even though H1 was formally posed against WIG20 alone. All of this sits comfortably with the variance-reduction prediction of forecast-combination theory: combining heterogeneous directional forecasts damps the variance of the resulting equity curve, and the damping is visible in the volatility and in the drawdown path alike.

H2 fares similarly. Against the strongest of the five single-agent runs, `only_rsi` at 0.16, the full ensemble returns 0.23, an advantage of more than six percentage points over the test window. The paired bootstrap puts that advantage at 0.268, short of the $p < 0.05$ mark, which is unsurprising given how hard Sharpe-like statistics are to establish over windows shorter than 400 trading days [12]. Read through the combination argument, the sign of the gap is what matters: a diversified score carries less error variance than any of its constituents, and it converts that into appreciably more total return without a matching increase in drawdown.

H3 receives the firmest support of the three. Deleting any one signal agent leaves the test-window total return inside a narrow band about the full ensemble, and even the weakest leave-one-out variant still finishes above 0.19 on total return, above 1 on Sharpe, and above buy-and-hold on Sortino.

Two entries in the ablation warrant comment. Dropping MACD, and with it part of the momentum family, nudges the headline metrics upward, which says that the fixed prior allots momentum more weight than this particular period rewarded. Section 5.3.2 reaches the same verdict by a different route, since deliberately reducing the momentum weight also improves the window, and it adds the caveat that fitting the weights in sample does not reliably capture such period-specific adjustments out of sample. The Bollinger agent is the opposite case: under the selected configuration it opens very few trades at all, so removing it barely disturbs the ensemble, and it functions on this window as a largely dormant member of the forecast pool.

Together these results establish that no individual agent is load-bearing, which is what forecast-combination theory anticipates so long as two imperfectly correlated forecasters survive in the pool. Section 5.3.2 carries the same robustness across from the roster of agents to the weighting itself, since neither perturbing, nor reversing, nor re-estimating the family weights from data overturns the qualitative conclusion.

5.4.3. Cross-market generalisation

The obvious objection to everything above is that it describes a single sample. Section 5.3.7 answers that objection by rebuilding the study on the FTSE 100, where the constituent set, the currency and the cost structure all differ, the last of these including an asymmetric 50 bps tax levied on purchases only. The conclusion survives the change intact. Over a bull-dominated window the ensemble again finishes behind passive buy-and-hold on gross total return while posting a better Sharpe ratio (1.37 against 1.13), a shallower maximum drawdown (−0.07 against −0.13), and better Sortino and Ulcer figures, and it again passes the benchmark on total return once the two

series are placed on a common ex-ante risk budget.

Recovering the same trade-off twice, and doing so under an added stamp duty that falls hardest on the more active of the two strategies, argues that the drawdown and downside-risk advantage is structural to forecast combination rather than a coincidence of one exchange. The transfer is nonetheless imperfect. Correction days are almost absent from the FTSE window, which denies the correction-regime defence that does most of the work on WIG20 any room to operate, and the integrated-drawdown advantage thins until the Martin ratio tips marginally to buy-and-hold. That is the regime dependence the adaptive markets hypothesis would lead one to expect [44]: the ensemble earns its keep in the regimes it was built to defend, and a window that is bull almost throughout is the least hospitable setting it could be given.

Three further considerations bound how widely the findings should be read beyond these two exchanges. The first is that technical rules do not pay equally everywhere, and what they pay decays as a market matures. Survey work finds rule profitability concentrated in emerging and thinly traded venues and progressively harder to demonstrate in developed ones once transaction costs and data-snooping corrections are imposed [55], and earlier evidence across Asian markets ties the same profitability to how developed the market is and how large its trading frictions are [56].

Placed on that spectrum, the two markets studied here behave like moderately mature ones in which chasing a single signal has stopped paying, since every single-agent baseline comes out unprofitable or marginal and neither learned model attains a statistically significant edge, and yet in which a disciplined and diversified risk overlay continues to contribute on the risk axis. It follows that the gross-return comparison should look kinder to the ensemble in less efficient, higher-friction markets, whereas in highly developed ones the risk-control profile is the component that ought to travel.

The second consideration is that what transfers is the structure rather than the numbers. Agent interfaces, the anchored walk-forward procedure, the calibrated cost model and the whole evaluation battery carry over unaltered to any exchange that publishes dependable daily closes, and the FTSE replication needed nothing beyond a fresh data source and a fresh cost model. The calibrations themselves, meaning thresholds, weights, commissions and taxes, have to be re-derived market by market, as the asymmetric UK stamp duty makes plain.

The third is that the contribution is regime-bound. It accrues in corrections and sideways phases, so whatever advantage the ensemble realises on a given market will follow the regime composition of the window it is measured over. Sustained bull regimes reward passive exposure mechanically, as both test windows examined here demonstrate. Charting where the framework does and does not hold across a wider set of developed and emerging indices, as Section 6.2 proposes, is accordingly the most direct continuation of this study.

5.4.4. Why gross total return understates the contribution of the ensemble

Convention compares a trading strategy with buy-and-hold by setting one gross total return beside another and awarding the verdict to whichever is larger. Applied to the present test window, that convention hands the verdict to the passive benchmark, and it does so while the ensemble is running at under half the benchmark drawdown and ahead of it on the Sharpe, Sortino, Calmar, Ulcer and Martin measures alike.

Three observations indicate that a single gross number is the wrong summary statistic in this setting. The first is that being out of the market is a design feature rather than a failure: the decision-agent thresholds withhold a position until the combined score clears a confidence

level, so across a window that is 52.87% bull, any partially invested strategy trails a fully invested one on gross cumulative return more or less by construction. The second is the volatility-matched result of Section 5.2.4, which shows that once the two series are given the same ex-ante risk budget the ensemble returns +18.55 percentage points more than buy-and-hold, and still at a smaller absolute drawdown. The third is the regime decomposition of Section 5.2.5, which locates the entire aggregate shortfall inside the bull regime, leaving the ensemble level with or ahead of buy-and-hold on gross return through the correction and sideways regimes.

Taken together, the three reframe what is being compared. The ensemble contributes as a *risk-aware decision support tool*, one that gives up a portion of bull-regime upside in exchange for shallower drawdowns and better risk-adjusted performance across the window as a whole.

5.4.5. *Connections to the broader empirical literature*

Three well-established regularities of empirical finance align with the pattern documented in Sections 5.2 and 5.3.

The first concerns momentum. Time-series and cross-sectional momentum have been shown to be empirically orthogonal, so that portfolios holding both achieve better risk-adjusted results than portfolios holding either [31, 33, 32]. What Table 5.7 records between the momentum family (MACD, ROC) and the mean-reversion family (RSI) is that same phenomenon observed one level down, at the signal rather than at the portfolio: the two families fall into disagreement precisely on the occasions when one of them would have been wrong unaided, and the decision-agent score folds the disagreement into a more cautious directional bet than either family would have taken by itself.

The second is the adaptive markets hypothesis [44], on which no individual trading rule can prevail across every regime, for the reason that the efficiency of a market with respect to any particular rule varies over time. Section 5.2.5 bears this out. A trend-weighted ensemble cedes ground through the bull regime, where a passive benchmark mechanically outpaces anything less than fully invested, and recovers it through corrections and sideways periods, where the mean-reversion and volatility agents supply directional information that the trend agent alone would not see.

The third comes from the measurement literature, which has argued for decades that the Sharpe ratio on its own cannot properly rank strategies whose drawdown profiles differ [21, 20, 22]. The gap between the two advantages measured here makes the argument concrete: on Sharpe the ensemble leads by a factor of 1.3 (1.06 against 0.82), on the Martin ratio by a factor of 2.1 (5.29 against 2.49). Adjusting for realised volatility and adjusting for the full drawdown path are evidently not the same adjustment, which is why the headline claim of this thesis is stated in terms of Calmar and Ulcer rather than of Sharpe alone.

5.4.6. *Practical implications for a retail investor*

Two implications matter for a retail investor working on the WSE with ordinary daily data and typical Polish brokerage terms. The first is the drawdown result. The worst realised loss is cut by better than a factor of two, and the integrated drawdown captured by the Ulcer Index by close to two-thirds. That axis tends to govern how retail portfolios are judged in practice, partly because maximum drawdown carries a direct operational meaning, being the deepest unrealised loss the investor has to sit through, and partly because extended drawdowns are empirically associated with abandoning the plan originally adopted.

The second is durability under costs. Every plausible pairing of commission and slippage in the sweep leaves the risk-adjusted advantage over buy-and-hold intact, the punitive 100 bps and 20 bps corner included, which sits well above what WSE retail tariffs currently charge. Neither implication requires leverage, and both are usable in the form of decision-support signals laid over a passive benchmark rather than as a substitute for holding it.

Two operational points follow. One is cadence. Nothing in the five agents or the decision agent consumes anything finer than an end-of-day close, so deploying the system amounts to running the pipeline once the WSE has closed and acting on what it recommends at the open or the close of the following session. That requires no intraday infrastructure, no tick data and no order-book integration, which puts adoption within reach outside the institutional segment.

The other is monitoring. Linearity of the decision-agent score in (3.5), weighted as it is by confidence and by family, means the score can be broken back down into per-agent contributions on any given day and charted next to the equity curve. The investor obtains from this an interpretable record of which agents moved each position, and a straightforward way of noticing structural shifts in the signal distribution, an agent whose confidence collapses for a prolonged stretch being the obvious case, with no retraining involved.

Interpretability of that kind distinguishes a theory-motivated forecast-combination design from end-to-end deep-reinforcement-learning trading systems, which as a rule emit a scalar action carrying no transparent attribution back to the individual input signals.

On the markets examined here that transparency costs very little in performance. The WIG20 comparison puts the ensemble ahead of both learned baselines of Section 5.1.5, structurally unlike each other though they are, fed identical price history and charged identical costs, across total return and every risk-adjusted ratio, the one exception being the marginally shallower GBM drawdown that follows from its taking almost no risk at all (Section 5.3.3). The FTSE 100 window reverses the point estimates, and the total-return gap of the GBM is marginally significant there, but the learned models reach those figures by staying close to fully invested through a window that is bull almost throughout, which leaves them statistically indistinguishable from passive buy-and-hold and without the drawdown control that motivates the ensemble in the first place (Section 5.3.7). Full per-signal attribution is therefore retained at no loss of competitiveness on the risk axis against opaque learned models.

5.4.7. *Deployment and scalability*

Whether the system can leave the backtest turns on a handful of practical properties, and on most of them the transparent design is favourably placed.

Computational cost. Nothing at decision time iterates. Every signal agent evaluates a closed-form indicator over a rolling window, updated incrementally for each symbol, and the decision agent collapses the per-symbol signals into a single confidence-weighted sum followed by a threshold test, exactly as (3.5) states. No training, no gradient computation and no specialised hardware enter the decision path, and one pass over the closing prices of the day on a commodity CPU is enough. The LSTM and GBM baselines of Section 5.1.5 do not share this property, nor do end-to-end deep-reinforcement-learning systems, all of which have to be trained and then retrained periodically.

Scalability across symbols and markets. Daily work scales as $O(N_{\text{symbols}} \times N_{\text{agents}})$, linear in the size of the traded universe, and it parallelises trivially, since each agent reads the history of one symbol on its own through the shared base-agent interface. Section 5.3.7 supplies

concrete evidence of market portability: aiming the whole pipeline at another exchange proved to be a matter of configuration, a new data source and a new cost model, and touched neither the agents nor the decision layer.

Liquidity. Only large-capitalisation index constituents are traded, at the daily close, and the selected configuration turns over slowly, so at retail size the resting-order footprint stays small. Price impact, partial fills and order-book dynamics fall outside the model and are treated as a limitation in Section 6.2.

Execution latency. Signals are produced once the session has closed and acted on in the next one, so delays measured in seconds or minutes are immaterial, and neither co-location nor tick data nor order-book integration is called for. On the spectrum of trading infrastructure this places the design at the opposite end from high-frequency trading.

Portfolio scalability and the retail-institutional distinction. Capital scales mechanically under the fractional position-sizing rule, but the assumption that orders fill at the closing price without impact caps the deployable capital against average daily volume, which makes the profile most defensible at retail-to-moderate assets under management. A retail investor can run the system as an end-of-day decision-support overlay on a passive benchmark, with neither leverage nor dedicated infrastructure. An institutional user would find the linear scaling and the market-agnostic interface adequate for wider universes, but would at that point be bound by the need for explicit price-impact modelling and by the analytical-only leverage caveat of Section 5.2.4.

5.4.8. Design implications

The experiments carry three implications for the design. The first is that the transparency of the decision-agent score, linear in the agent signals and weighted by confidence and by family, serves theory and practice at once. The contribution each agent makes on a given day is legible directly from the score, and the ablation shows that this attribution agrees with the marginal contribution of that agent to the headline metrics.

The second is that Table 5.6 holds the Sharpe ratio strictly above the passive benchmark at every plausible cost configuration, which indicates that the ensemble is not being kept alive by a favourable cost calibration.

The third is that the selected (mode, $c_{\min}$) pair holds steady across all three walk-forward folds, which suggests that a plain grid search over a long enough training window suffices, and that heavier parameter-adaptation machinery earns nothing at daily frequency on WIG20.

5.5. Chapter summary

The evaluation reported in this chapter was designed to produce results that survive scrutiny rather than results that look impressive. Parameters were selected on a training window and never revisited, every trade was charged realistic commissions and slippage, and every headline figure was accompanied by a bootstrap confidence interval and a paired significance test.

Under that protocol, the finding is consistent and narrow. The ensemble does not beat passive buy-and-hold on gross total return in either test window, and on the WIG20 panel it trails by eight percentage points. It does, however, lead on every risk-adjusted metric considered, halves the maximum drawdown, reduces the integrated drawdown by nearly two-thirds, and overtakes the benchmark on total return once the two series are placed on the same ex-ante risk budget. The regime decomposition locates the source of this profile precisely: the ensemble defends during corrections and carries during sideways phases, and pays for both by under-participating in bull

legs.

The robustness checks constrain how much of this can be attributed to chance or to arbitrary design choices. The profile survives the removal of any single agent, perturbation and re-estimation of the family weights, every plausible transaction-cost calibration, a change of the selection objective, a three-fold chronological re-split, and a full replication on a second market with a different cost regime. What it does not survive is the demand for conventional statistical significance on a single sub-two-year window, and that limitation is stated plainly rather than glossed over.

One result deserves emphasis because it cuts against the original design intent. The confidence values that the decision agent uses as weights carry no measurable information about signal-level accuracy. The system works, on the risk axis, but not for the reason its architecture suggests: what operates is the pooling of signals that correlate weakly or negatively across the methodological families, not precision weighting. Chapter 6 returns to what this implies for the contribution of the thesis and for the work that should follow it.

6. SUMMARY

The aim of this thesis, stated in Section 1.2, was to design a transparent multi-agent decision support system for stock trading on the Warsaw Stock Exchange and to evaluate it under a protocol rigorous enough that its results can be trusted. Both parts of that aim were met, and the second turned out to matter more than the first.

The system itself is deliberately simple: five signal agents, each confined to one methodological family, feed a decision agent that combines their signals into a linear, confidence-weighted score and thresholds it into a trading action. The evaluation around it is not simple. It combines anchored walk-forward tuning, calibrated commissions and slippage, five risk-adjusted metrics, a volatility-matched comparison, bootstrap confidence intervals and paired tests, a decomposition into market regimes, a leave-one-agent-out ablation, sensitivity analyses over the family weights and the selection objective, a cost-sensitivity sweep, an audit of the confidence heuristic, two machine-learning baselines, and a full replication on a second market.

What that evaluation established is narrower than what the project set out to find, and more defensible. The ensemble is not a return enhancer. Passive buy-and-hold earned more on both test windows, because both were dominated by bull regimes and a partially invested strategy cannot keep pace with a fully invested one in such conditions. The ensemble is, however, a risk reducer, and consistently so: it halved the maximum drawdown on WIG20, cut the integrated drawdown by nearly two-thirds, led on every risk-adjusted ratio reported, and overtook the benchmark on total return once both series were placed on the same ex-ante risk budget. Section 3.3 stated three hypotheses; the evidence is consistent with all three on point estimates, and with none of them at conventional levels of statistical significance on a single window of under two years.

6.1. Conclusions

The work supports the following conclusions:

1. the multi-agent system improves risk-adjusted performance rather than gross return, leading buy-and-hold on Sharpe, Sortino, Calmar, Ulcer, and Martin measures while trailing it on cumulative profit in bull-dominated windows,
2. the advantage is concentrated where the design predicts it should be, namely +12.57 percentage points of correction-regime defence and +5.14 percentage points of sideways-regime carry, paid for by underperformance in the bull phase,
3. the profile is not an artefact of any single design choice, since it survives the removal of any one agent, perturbation and re-estimation of the family weights, every plausible cost calibration, a change of selection objective, and a three-fold chronological re-split,
4. the profile is not specific to the Warsaw Stock Exchange, since the FTSE 100 replication reproduces it under a different constituent set, currency, and cost regime that includes an asymmetric buy-side transaction tax,
5. the transparent ensemble remains competitive with an LSTM and a gradient-boosted-trees classifier evaluated under the identical protocol, both of which achieve their apparent edge on the FTSE window by remaining near-fully invested through a bull market,

6. the confidence heuristic does not work as designed, since confidence values carry no measurable information about signal-level accuracy on either market, so the inverse-variance reading of the aggregation rule is a conceptual analogy rather than a calibrated mechanism,
7. the mechanism that does operate is error diversity, since pooling signals that correlate weakly or negatively across the methodological families lowers the variance of the aggregate position however the pool is weighted,
8. none of the observed advantages reaches conventional statistical significance on a single sub-two-year test window, and they are therefore reported as economically meaningful rather than as statistically established.

The sixth and seventh conclusions are the ones that were not anticipated at the outset, and they are worth stating plainly. The architecture was motivated by the idea that a confidence value could act as a proxy for the reliability of an agent at a given moment, so that the decision agent would perform something close to precision-weighted averaging. The confidence-validation experiment of Section 5.2.8 shows that this is not what happens. The system works on the risk axis, but for the weaker and more robust of the two reasons that forecast-combination theory offers.

Methodologically, the evaluation pipeline is the more transferable contribution. It closes the four gaps identified in Chapter 2 and transfers as a template to comparable studies on smaller equity markets, where preserving capital through turbulent periods may count for more than pursuing the next bull leg.

6.2. Difficulties and limitations

The study carries several limitations.

First, the two markets evaluated are both blue-chip indices of large capitalisation, sampled at daily frequency. Extending the finding to thinner and less liquid stocks, to other asset classes or to intraday data would demand a separate evaluation, and agents built around daily closes would need adapting before any intraday use.

Second, the cost model covers commissions and slippage and is stress-tested across a wide grid, but liquidity constraints, partial fills and the price impact of larger orders all lie outside it. Sizeable orders would move the closing price and would call for simulation at the level of the order book.

Third, nothing beyond price enters the system. Fundamental, news and alternative sources are known to carry information orthogonal to price-based indicators, and each would sit naturally in the forecast-combination framework as a further agent with its own error structure.

Fourth, the decision agent runs on fixed family weights and heuristic confidence values. Section 5.3.2 finds the headline conclusions robust to perturbing, reordering and re-estimating those weights, and finds in-sample optimisation no reliable improvement on the fixed prior; the weights nonetheless remain static over time, and regime-conditional weighting is untested. Section 5.2.8 finds the confidence values uninformative about signal-level accuracy, so no calibrated precision weighting can be claimed and the link to classical forecast-combination theory stays conceptual.

Fifth, the test window reaches across bull, correction and sideways regimes but remains a single window of under two years, on which the bootstrap carries limited power against moderate effect sizes. Point estimates and confidence intervals consequently bear as much of the argument here as p -values do.

Sixth, matching the two series on volatility calls for leverage of $\lambda = 2.06$, which lies beyond the reach of a typical retail investor on the WSE. That comparison is a counterfactual on risk budget and not an operational recommendation.

Finally, no live-trading exercise was undertaken, and any claim about realised performance in production would require one.

Two difficulties beyond these shaped the course of the work. One was computational. Running the full battery means re-running the backtest hundreds of times over grid points, ablations, weight perturbations and cost configurations, which stayed intractable until signal computation was separated from portfolio simulation and the agent signals precomputed once (Algorithm 1). Runtime then fell by better than an order of magnitude, and the sensitivity analyses became affordable.

The other was interpretive. An early version judged the system against buy-and-hold on cumulative return alone, and by that measure it looked like a failure. Seeing that the comparison was confounded, since a partially invested strategy was being set against a fully invested benchmark running at roughly twice the volatility, is what prompted the risk-adjusted battery, the regime decomposition and the volatility-matched comparison that now sit at the core of the evaluation.

6.3. The way forward

Each limitation points to a direction for further work.

The FTSE 100 validation invites extension to further indices, the German DAX, the Italian FTSE MIB or the US S&P 500 among them, and to smaller-capitalisation and emerging-market segments, which together would map more precisely the conditions under which the risk-adjusted advantage holds. Moving to intraday frequencies would require revising the signal agents and the transaction-cost model together, and would expose the system to market-microstructure effects it currently ignores [57].

Fundamental and alternative data would enter as further signal agents behind the same base-agent interface, which the modular design of Chapter 4 makes straightforward.

Since neither static in-sample optimisation nor inverse-variance estimation adds much (Section 5.3.2), the promising weighting extension is a regime-conditional one, raising the trend family through bull regimes and lowering it through corrections. That follows naturally from the present fixed-weight design and should recover part of the bull-regime shortfall that Section 5.2.5 identifies.

A complementary direction replaces the heuristic confidence values with calibrated probabilities, mapping indicator magnitudes onto empirical hit rates estimated on the training window by an isotonic or Platt-type fit [54]. Genuine precision weighting would then become available to the decision layer, and the conceptual link to forecast-combination theory would turn into a quantitative one open to test. Given what Section 5.2.8 found, this is the single most valuable extension of the present design.

The roster itself could be made adaptive rather than fixed, its active pool selected per regime or per symbol from recent forecast accuracy, so that an agent contributes only where it has demonstrated skill. More ambitiously still, a learning-based ensembling layer, a stacked meta-learner or a gating network fitted to the data, could displace the linear confidence-weighted score. The difficulty there is to admit a richer and possibly non-linear combiner without surrendering the per-agent attribution and interpretability that motivate the present design.

Finally, a walk-forward exercise against a paper-trading broker interface would close the gap between backtest and realistic execution, and would put to the test the operational assumptions

the backtest takes for granted, chiefly end-of-day fills at closing prices without market impact.

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